

Molecular characterization of gustatory second-order neurons reveals integrative mechanisms of gustatory and metabolic information

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This study provides **valuable** insights into the organization of second-order circuits of gustatory neurons, particularly in how these circuits integrate opposing taste inputs and are modulated by metabolic state to regulate feeding behavior. Through an elegant combination of complementary techniques, the authors identify the target neurons involved in gustatory integration. The evidence supporting their conclusions is **convincing**.

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Abstract

Animals must balance the urgent need to find food during starvation with the critical necessity to avoid toxic substances to ensure their survival. In *Drosophila*, specialized Gustatory Receptors (GRs) expressed in Gustatory Receptor Neurons (GRNs) are critical for distinguishing between nutritious and potentially toxic food. GRNs project their axons from taste organs to the Subesophageal Zone (SEZ) in the Central Brain (CB) of *Drosophila*, where gustatory information is processed. Although the roles of GRs and GRNs are well-documented, the processing of gustatory information in the SEZ remains unclear. To better understand gustatory sensory processing and feeding decision-making, we molecularly characterized the first layer of gustatory interneurons, referred to as Gustatory Second-Order Neurons (G2Ns), which receive direct input from GRNs. Using trans-synaptic tracing with *trans*-Tango, cell sorting, and bulk RNAseq under fed and starved conditions, we discovered that G2Ns vary based on gustatory input and that their molecular profile changes with the fly's metabolic state. Further data analysis has revealed that a pair of neurons in the SEZ, expressing the neuropeptide Leucokinin (SELK neurons), receive simultaneous input from GRNs sensing bitter (potentially toxic) and sweet (nutritious) information. Additionally, these neurons also receive inputs regarding the starvation levels of the fly. These results highlight a novel mechanism of feeding regulation and metabolic integration.

Introduction

Animals face constant dangers in their environment that they must avoid to survive, being starvation and poisoning two critical threats. To prevent death by starvation, animals must continually seek food, even if it means risking consumption of potentially harmful substances^{1,2,3}. Therefore, accepting or rejecting food potentially poisonous will depend on the general nutritious content and toxicity of the meal, together with the starvation level of the animal. Similar to other animals, *Drosophila melanogaster* can discern different taste qualities (sweet, bitter, salty, sour, umami or carbonation) through various populations of Gustatory Receptor Neurons (GRNs) tuned to each of these qualities⁴. GRNs are distributed across the *Drosophila*'s body including the proboscis, legs, wing margins, and the ovipositor organ, expressing a combination of receptors for detecting various food compounds⁵. There are four major families of receptors involved in gustatory perception: Gustatory Receptors (GRs), Ionotropic Receptors (IRs), Transient Receptor Proteins (TRPs), and pickpocket (ppk) channels⁶. Specific GRNs are associated with particular valences, such as sweet or bitter tastes. For instance, Gr64f is a receptor for sugar detection promoting feeding behavior^{7,8}, while Gr66a is present in GRNs that detect bitter compounds leading to feeding rejection^{9,10}.

GRNs project their axons to the Subesophageal Zone (SEZ), the main brain region in flies for integrating and processing gustatory information. Unlike the olfactory system, where olfactory sensory neurons expressing the same receptors converge in the same glomeruli to synapse with projection neurons, the gustatory system lacks such anatomical subdivisions^{5,11}. However, neurons expressing receptors that respond to sweet compounds and elicit attractive behaviors converge in specific SEZ regions without overlapping neurons detecting bitter compounds and inducing feeding rejection^{10,12}.

The development of extensive collections of Gal4 lines¹³, split-Gal4 collections¹⁴, and databases for searchable neurons¹⁵, has facilitated the identification of gustatory interneurons in the SEZ involved in feeding behavior. Those databases have been used to design experimental procedures that has help to the identification of some gustatory interneurons involved in integrating gustatory information and inducing feeding^{1,16-22}. These studies have identified command neurons critical for feeding behavior, some directly receiving input from GRNs, such as certain GABAergic neurons¹, while others, like the Feeding neuron (Fdg), act as a hub to control food acceptance without being connected directly to any GRN¹⁶. Despite these advances, the role of gustatory interneurons in integrating gustatory information and the metabolic state of the fly remains largely unexplored. The recent development of the *Drosophila* brain connectome through electron microscopy is advancing our understanding of SEZ connectivity and the complex interactions among GRNs^{23,24}.

We have aimed to elucidate the integration of gustatory information and the metabolic state, by analyzing the first layer of interneurons that collect gustatory information, Gustatory Second-Order Neurons (G2Ns)²³. We have decided to focus on understanding how flies integrate gustatory information that codes for opposing behaviors: sweet tastants elicit feeding whereas bitter compounds induce food rejection¹¹. We have followed a molecular approach to identify G2Ns. First, we employed the *trans*-Tango genetic tool²⁵, which labels postsynaptic neurons connected to specific presynaptic neurons. Later, using Fluorescent Activated Cell Sorting (FACS) of labeled G2Ns and bulk RNA sequencing (RNAseq), we characterized the molecular profiles of postsynaptic neurons connected to sweet and bitter GRNs in fed and starved conditions for the first time. These analyses show that G2Ns receive input from molecularly distinct GRNs, and their transcriptional profiles change upon starvation, presumably to adapt to the new metabolic state. Using molecular and connectomic techniques, we further demonstrate that the two *Leucokinin*-expressing neurons in the SEZ (SELK neurons) are G2Ns that simultaneously receive sweet and

bitter GRN input. Behavioral experiments revealed that Leucokinin neurons integrate information regarding the metabolic state of the fly and are involved in the processing of bitter and sweet information that impacts the initiation of feeding behavior.

Results

Molecular characterization of gustatory second-order neurons

We employed the trans-synaptic tracing method *trans*-Tango²⁵, to label G2Ns synaptically connected to sweet and bitter GRNs. To label G2Ns receiving sweet information, we used the *Gr64f-Gal4* transgene^{7,10}, and for G2Ns receiving bitter information, we used the *Gr66a-Gal4* transgene^{10,26}. These Gal4 transgenes enabled us to label multiple G2Ns of interest (Sup. Figure 1.1A and B). The number of postsynaptic G2Ns differed between the GRNs, indicating differences in information processing for different tastants (Sup. Figure 1.1C and D). Since satiety and hunger significantly influence behavior, modifying olfactory behavior²⁷, memory formation²⁸ and food consumption^{2,29}, we extended our analysis to characterize molecularly G2Ns receiving different gustatory inputs in two metabolic states: fed and 24-hour starved flies. Given that all G2Ns are fluorescently labeled with mtdTomato, we used FACS to separate and collect the cells for bitter and sweet G2N populations (*Gr66a*^{G2Ns} and *Gr64f*^{G2Ns}, respectively) in the two metabolic conditions (see Materials and Methods for a detailed description of the sorting procedure and sample number in Table 1). We then performed bulk RNAseq for each condition (Figure 1A).

The data's Principal Component Analysis (PCA) revealed that the transcriptomic profiles of the two G2N populations are different and that the metabolic state (fed vs. starved) significantly influences gene expression without affecting the separation among the populations in the PC space (Figure 1B). These findings suggest that the first layer of neurons collecting sensory information is already modulated by the fly's metabolic state, potentially impacting subsequent information processing.

We further analyzed the changes in gene expression between fed and starved conditions across the two G2N populations. Many genes exhibited differential expression in starved flies compared to fed flies (Figure 1C). Gene Ontology (GO) analysis indicated that several genes were involved in nervous system development and synapse organization, suggesting the importance of synapse remodeling during starvation adaptation. Additionally, GO terms related to the response to sensory stimuli and stress response were identified, implying that starvation induces molecular signatures of stress in flies. Finally, some genes associated with signaling pathways indicated that neurotransmitter and neuropeptide signaling processes might be altered during starvation, leading to specific adaptations in the neuronal circuits involving G2Ns (Sup. Figure 1.2).

Characterization of neurotransmitter and neuropeptide expression

Next, we focused on the expression of neurotransmitters, neuropeptides, and their respective receptors as they are known to modulate feeding among many other behaviors³⁰. For example, the CCHamide 1 receptor regulates starvation-induced olfactory modulation³¹; NPF (Neuropeptide F) modulates taste perception in starved flies by increasing sensitivity to sugar and decreasing sensitivity to bitter by promoting GABAergic inhibition onto bitter GRNs²; starvation enhances behavioral sugar-sensitivity via increased dopamine release onto sweet GRNs, indirectly decreasing sensitivity to bitter tastants³²; and octopamine signaling affects bitter signaling in starved flies³.

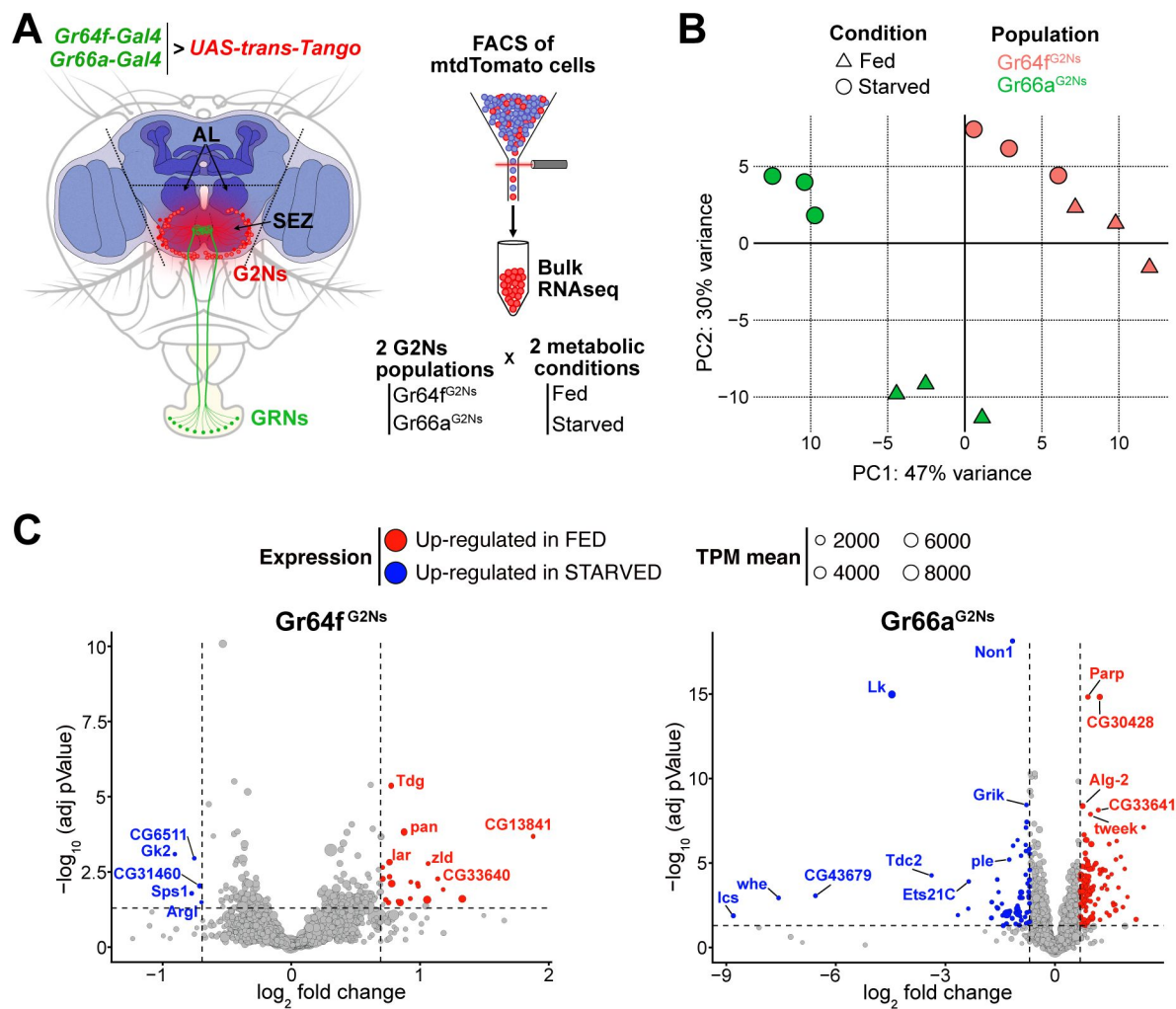


Figure 1.

The metabolic state affects the gene profiles of the Gustatory Second-Order Neurons (G2Ns) populations analyzed.

(A) The scheme depicts the region of the brain dissected and the sorting of the G2Ns according to the input. Two populations of *trans-Tango* neurons were labeled using *Gr64f-Gal4* and *Gr66a-Gal4* transgenic lines. Flies were either fed or food-starved for 24 hours. (B) Principal Component Analysis (PCA) of the genes. (C) Volcano plots show the genes up- and down-regulated in fed and starved conditions in the two populations of G2Ns.

Fed Flies

Replicate	#Brains dissected	#Cells collected
Gr64f_1	406	21017
Gr64f_2	512	23782
Gr64f_3	191	21428
Gr66a_1	519	21260
Gr66a_2	531	21599
Gr66a_3	594	21433

Starved Flies

Replicate	#Brains dissected	#Cells collected
Gr64f_1	372	23126
Gr64f_2	315	23387
Gr64f_3	459	23267
Gr66a_1	657	27555
Gr66a_2	718	27090
Gr66a_3	636	26991

Table 1.

Number of brains and cells dissected

Our RNAseq data revealed that the most prominent neurotransmitter receptor expressed in all G2Ns was *nAChRbeta1* (nicotinic Acetylcholine Receptor Beta 1) (**Figure 2A**). This is consistent with previous findings showing that GRNs are mostly cholinergic. Other acetylcholine and neurotransmitter receptors were also expressed in G2Ns. This indicates that while G2Ns receive significant input from GRNs, they also integrate information from a complex network of other neurons using different neurotransmitters. We found no variation in expression levels for any neurotransmitter receptors when comparing fed and starved conditions. Regarding neurotransmitter expression, *Gad1* (Glutamic acid decarboxylase 1) and *Ddc* (Dopa decarboxylase), two essential enzymes involved in the synthesis of GABA and dopamine, respectively, were highly expressed in the two G2N populations for both metabolic states (**Figure 2B**). Other enzymes involved in the GABA pathway, like *VGAT* (vesicular GABA transporter) or *GABAT* (γ-aminobutyric acid transaminase), were also expressed but at lower levels than *Gad1*. Similarly, other enzymes or transporters related to other neurotransmitters, like glutamate (Glu) or serotonin (5-HT), were present but at lower levels.

Regarding neuropeptide receptors, we found many receptors expressed in all G2Ns populations with varying expression levels (**Figure 2C**). For instance, the *EcR* (Ecdysone Receptor) showed expression in all three populations with a slight decrease under starvation, indicating a possible role in adapting to food deprivation. Additionally, *sNPF-R* (short Neuropeptide F Receptor) was highly expressed with similar expression patterns as its ligand sNPF (short Neuropeptide F), highlighting the importance of sNPF and its signaling pathway in regulating starvation and feeding.

Neuropeptides showed variable expression patterns between populations and conditions (**Figure 2D**). For example, *sNPF* was highly expressed in all G2Ns, particularly in the Gr66a^{G2N} population, without changes associated with the metabolic state. sNPF production, which is related to circulating sucrose levels, impacts food intake by integrating sweet and bitter information in food-deprived flies. This highlights the role of sNPF and its receptor (sNPF-R) in regulating the feeding behavior according to the metabolic state, as it is secreted in the midgut and by neurosecretory cells in the brain. Dh31 (Diuretic hormone 31) and Dh44 (Diuretic hormone 44), which are involved in desiccation tolerance and starvation adaptation, also showed notable expression in both G2N populations under starved and fed conditions. Nplp1 (Neuropeptide-like precursor 1), which is associated with synaptic organization in the nervous system and water balance, was also expressed in both G2Ns. Among all neuropeptides, however, Leucokinin (Lk) displayed the most striking expression change. Though *Lk* has no (or very low) expression in the Gr64f^{G2N} population under fed and starved conditions, and no (or very low) expression in the Gr66a^{G2N} under fed conditions, *Lk* showed significantly elevated expression in the Gr66a^{G2N} population under starved conditions. This expression pattern specific to Gr66a^{G2Ns} suggests that Lk may play a role in processing bitter information under starvation. We decided to explore this neuropeptide further to understand its role in integrating feeding behavior and the metabolic state of the flies.

Leucokinin expression is upregulated SELK neurons in the starved condition

Leucokinin is a neuropeptide with multiple roles in various physiological and behavioral processes. Three different Lk neuron groups exist in the adult *Drosophila* Central Nervous System (CNS). There are approximately 20 Lk neurons in the Ventral Nerve Cord (ABLKs), while in the CB, there are four Lk neurons, two located in the Lateral Horn (LHLK) and another two located in the SEZ (SELK). ABLKs use Lk as a hormonal signal targeting peripheral tissues, including renal tubules, to regulate water and ion homeostasis in response to desiccation, starvation, and ion stress, as Lk regulates fluid secretion in the Malpighian (renal) tubules. LHLK neurons are part of the output circuitry of the circadian clock, regulating locomotor activity and sleep suppression induced by starvation. Functional imaging revealed that LHLK

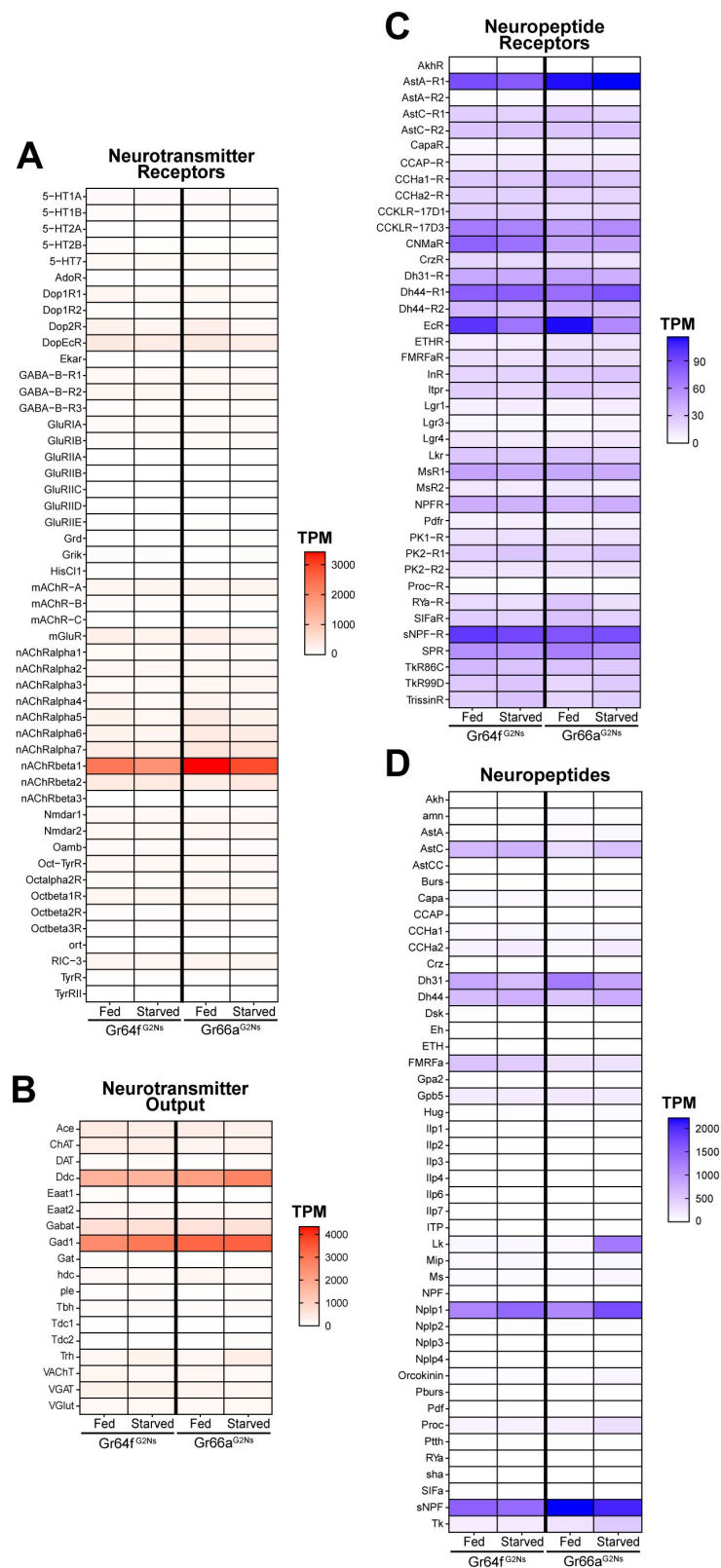


Figure 2.

Transcripts per Million (TPM) for Neuropeptides and Neurotransmitters.

TPMs for Neurotransmitters (A), Neurotransmitter Receptors (B), Neuropeptides (C) and Neuropeptide Receptors (D) for the two G2N populations in the two metabolic conditions studied (fed and 24h starved).

neurons, not SELK neurons, were required to suppress sleeping during starvation⁴⁴. It has been suggested that SELK neurons may regulate feeding as possible synaptic partners to the premotor feeding program. However, it is still not clear the role of SELK neurons in feeding behavior^{38,40}.

Due to the dissection method used for our RNAseq experiment (**Figure 1A**), we primarily captured SELK neurons rather than LHLK neurons. To confirm that starvation induces changes in *Lk* gene expression, we performed qPCR on brains from fed and starved *Drosophila*. Given that Lk neurons constitute 4 out of approximately 200,000 neurons in the *Drosophila* brain⁴⁵, we enriched our samples by expressing GFP in Lk neurons using the driver *Lk-Gal4*⁴⁶ combined with *UAS-mCD8::GFP* transgene and FACS of the GFP⁺ cells. We conducted qPCR analysis on the whole CB (including both LHLK and SELK neurons) as well as on the SEZ alone (containing only the 2 SELK neurons) in fed and starved conditions (**Figure 3A** and **3B**). Our results showed that starvation increased the expression level of *Lk* mRNA in the CB and also in the SELK neurons, indicating that *Lk* expression is indeed affected by the animal's starvation state (**Figure 3C**). To confirm that the increased mRNA expression translates into higher protein levels, we performed immunohistochemistry using an Lk antibody in fed and 24-hour starved wild-type flies. We quantified the fluorescence in the LHLK and SELK neurons and found that *Lk* expression was elevated under starvation conditions (**Figure 3D-F**) further validating our RNAseq results. Together, these three lines of evidence indicate that SELK neurons increase the expression of the Lk neuropeptide in 24-hour starved flies.

Leucokinin neurons are gustatory second-order neurons to GRNs

Our RNAseq data showed that the two SELK neurons receive almost exclusive input from bitter, but not sweet GRNs and that the metabolic state of the fly modulates the expression of *Lk*. To confirm that Lk neurons are indeed postsynaptic partners of bitter GRNs, we used an Lk antibody³⁹ to test the co-localization of Lk protein with the cells labeled in *GRNs-Gal4 > trans-Tango* flies. These experiments demonstrated that Lk signal colocalizes with trans-Tango in *Gr66a-Gal4 > trans-Tango* flies, confirming our expectations based on the RNAseq data (**Figure 4A**). Unexpectedly, however, we also found co-localization of the Lk antibody signal in *Gr64f-Gal4 > trans-Tango* flies, suggesting that SELK neurons do not receive exclusive input from bitter neurons but also from sweet-sensing neurons expressing the Gr64f receptor (**Figure 4B**).

To validate the synaptic connectivity between Gr64f^{GRNs} and Gr66a^{GRNs} with SELK neurons, we next used the GRASP (GFP Reconstruction Across Synaptic Partners) technique⁴⁷. GRASP can reveal the synaptic connectivity between two neurons, through the expression of half of the GFP protein in each candidate neuron. The reconstruction of the entire GFP protein in the synaptic cleft can be detected by a specific GFP antibody. We expressed half of the GFP (GFP¹⁻¹⁰) protein in the bitter GRNs using *Gr66a-LexA*⁴⁸ and sweet GRNs using *Gr64f-LexA*⁸ transgenes. The other half of GFP (GFP¹⁻¹⁰) protein was expressed in Lk neurons using the *Lk-Gal4* driver. GRASP results confirmed the synaptic connection between bitter GRNs and sweet GRN with SELK neurons in the SEZ (**Figure 5A** and **5C**), further confirming our previous *trans-Tango* result.

The GRASP technique can produce false positives, as a positive signal may result from the proximity of the neurons of interest. We decided to complement those results using active-GRASP⁴⁹. In this modified version, the presynaptic half of GFP (GFP¹⁻¹⁰) is fused to synaptobrevin. This fusion allows the GFP to be present in the membrane of the presynaptic neuron within the synaptic cleft when the presynaptic neuron is stimulated. In our experiments, we fed the flies with sucrose to activate sweet GRNs and caffeine to activate bitter GRNs. A positive signal in active-GRASP indicates synaptic connectivity between the candidate neurons and active communication between them. Our results with active-GRASP supported the findings from GRASP. Stimulation with sucrose resulted in clear positive GRASP signal in the *Gr64f-lexA, Lk-Gal4 > active-GRASP* flies, whereas stimulation with water did not produce a positive result (**Figure 5B**).

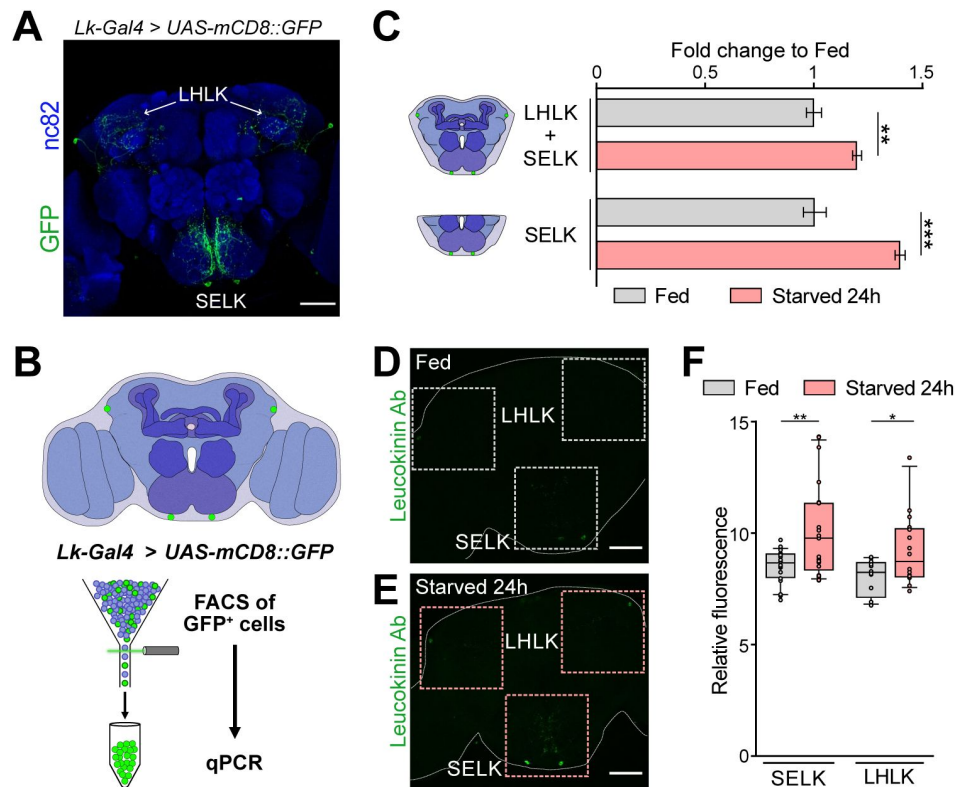


Figure 3.

Leucokinin expression is increased in starvation.

(A) Immunofluorescence with anti-GFP (green) and anti-nc82 (blue) on a whole-mount brain of a *Lk-Gal4 > UAS-mCD8::GFP*. (B) Scheme showing the approach used to select the Lk GFP⁺ neurons by FACS. (C) qPCR results for Lk neurons labeled with GFP and sorted by FACS. The central brain includes four Lk neurons, while the SEZ region includes only the two neurons located in this region. (D) Intensity fluorescence quantification in whole *Oregon-R* brains stained with anti-Lk antibody in fed and 24-hour starved flies. SELK Fed = 20 brains; SELK Starved = 21 brains; LHLK Fed = 12 brains; LHLK Starved = 16 brains. Statistical test: Wilcoxon rank sum test. **p*<0.05, ***p*<0.01, ****p*<0.0001. Scale bar = 50 μm.

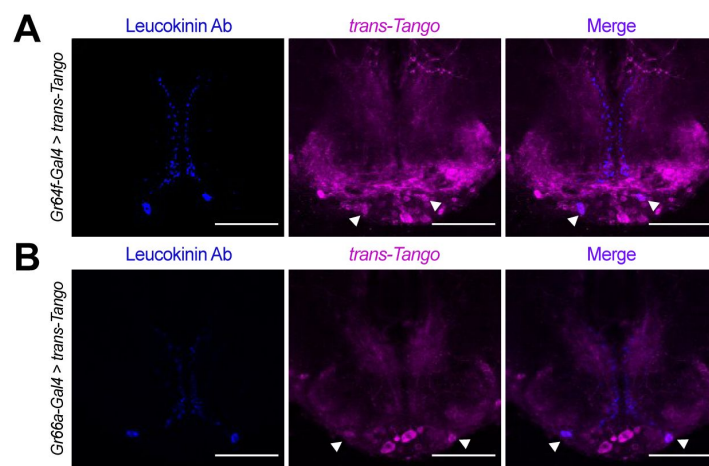


Figure 4.

Lk neurons co-localize with the *trans-Tango* signal.

Immunofluorescence with anti-Leucokinin (blue) and anti-RFP (magenta) on whole-mount brains of (A) *Gr64f-Gal4 > trans-Tango*, and (B) *Gr66a-Gal4 > trans-Tango*. Images correspond to the adult SEZ. Arrowheads point to SELK neurons. Scale bar = 50 μm .

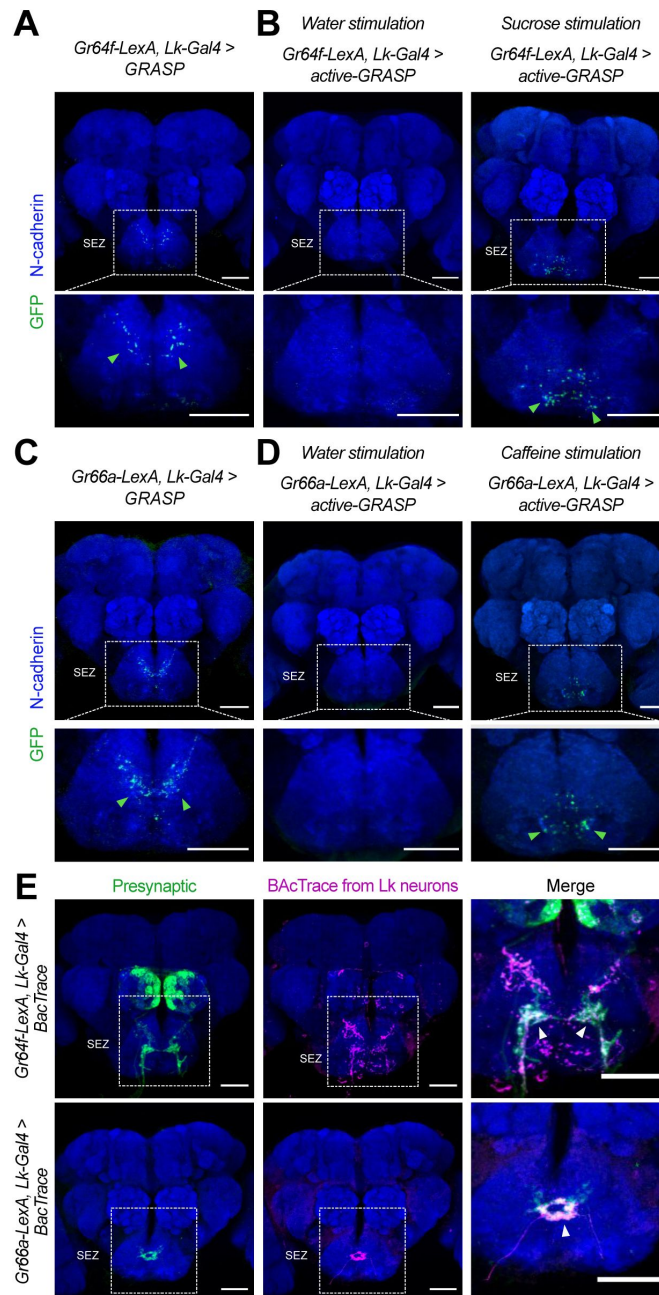


Figure 5.

Lk neurons are synaptically connected to Gr64f^{GRNs} and Gr66a^{GRNs}.

(A) **GRASP**: Immunofluorescence with anti-GFP (green) and anti-N-cadherin (blue) on whole-mount brains of *Gr64f-LexA > LexAop-CD4spGFP¹¹* and *Lk-Gal4 > UAS-CD4spGFP¹¹* flies. (B) **Active-GRASP**: Immunofluorescence with anti-GFP (green) and anti-N-cadherin (blue) on whole-mount brains of *Gr64f-LexA > LexAop-nSybspGFP¹¹* and *Lk-Gal4 > UAS-CD4spGFP¹¹* stimulated with water (left column) and sucrose (right column). (C) **GRASP**: Immunofluorescence with anti-GFP (green) and anti-N-cadherin (blue) on whole-mount brains of *Gr66a-LexA > LexAop-CD4spGFP¹¹* and *Lk-Gal4 > UAS-CD4spGFP¹¹*. (D) **Active-GRASP**: Immunofluorescence with anti-GFP (green) and anti-N-cadherin (blue) on whole-mount brains of *Gr66a-LexA > LexAop-nSybspGFP¹¹* and *Lk-Gal4 > UAS-CD4spGFP¹¹* stimulated with water (left column) and caffeine (right column). (E) Immunofluorescence with anti-GFP (green), anti-RFP (magenta), and anti-nc82 (blue) on whole-mount brains of *Gr64f-LexA, Lk-Gal4 > BacTrace* (top row) and *Gr66a-LexA, Lk-Gal4 > BacTrace* (bottom row). Arrowheads point to the BacTrace positive signal. Scale bar = 50 μm.

Similarly, caffeine stimulation produced a strong positive GRASP signal in the *Gr66a-LexA, Lk-Gal4* > *active-GRASP* flies, that was absent in the control group (water) (**Figure 5D**) (see Materials and Methods for a full description of the stimulation procedure). Our GRASP and active-GRASP results indicate that SELK neurons receive active input from sweet *Gr64f*^{GRNs} and bitter *Gr66a*^{GRNs}. Finally, we used BacTrace⁵⁰ to confirm that the connectivity between *Gr64f*^{GRNs} and *Gr66a*^{GRNs} neurons with SELK neurons is real. This technique works similarly to *trans*-Tango, but while *trans*-Tango labels postsynaptic neurons, BacTrace labels presynaptic partners. For our experiments, we used *Gr66a-LexA* and *Gr64f-LexA* as candidate presynaptic partners. Consistent with the GRASP and active-GRASP results, our BacTrace experiment indicated that sweet and bitter GRNs are presynaptic to SELK neurons (**Figure 5E**) further supporting a model in which Lk neurons are G2Ns receiving input from *Gr64f*^{GRNs} and *Gr66a*^{GRNs}. To our best knowledge, this is the first time a G2N has been identified as collecting information from two populations of GRNs that transduce sensory information of different valence, sweet-attractive and bitter-repulsive.

Connectivity of SELK neurons within the *Drosophila* brain

Using molecular techniques, we have demonstrated that SELK neurons are G2Ns for both bitter *Gr66a*^{GRNs} and sweet *Gr64f*^{GRNs}. We used the recently published Full Adult Fly Brain (FAFB) *Drosophila* connectome⁵¹ to identify the SELK neurons and further analyze their connectivity. As SELK neurons receive simultaneous input from sweet and bitter GRNs, we took advantage of recent research that identified groups of bitter and sweet-sensing GRNs and their projections to the SEZ in the *Drosophila* brain connectome (Sup. Figure 6A)²⁴. Using this dataset, we identify all postsynaptic partners to the *Gr64f*^{GRNs} and *Gr66a*^{GRNs} characterized, without limiting the number of synaptic contacts among neurons. We noted 12594 postsynaptic hits for *Gr66a*^{GRNs} and 36790 hits for *Gr64f*^{GRNs}. By intersecting both datasets, we obtained 333 downstream segment IDs as possible G2Ns receiving inputs from both GRNs. We refined our list to 17 candidates through visual inspection, focusing on neurons with projections within the SEZ (Sup. Figure 6B). To identify the SELK neuron among our candidates, we used the CMTK plugin in imageJ to register the immunohistochemistry images of the *Lk-Gal4* > *UAS-CD8::GFP* brains (**Figure 3A**) to a *Drosophila* reference brain (JRC2018U). Further, we skeletonized the images with ImageJ and uploaded them to the Flywire Gateway platform to compare the structure of the aligned SELK neurons with our candidates (Sup. Figure 6C). Via comparison with our candidates, we identified one candidate ID 720575940630808827 (GNG.685) (Sup. Figure 6D), which shared similar morphologies and projections to the right SELK neuron. The GNG.685 has a mirror twin, GNG.595, in the SEZ, and both are annotated as DNg68 in the Flywire database (**Figure 6A**). These candidates are classified as descending neurons, characteristic of known SELK neurons³⁸ and possibly also expressing the neurotransmitter acetylcholine. We consider the neurons DNg68 are the strongest candidates for representing SELK neurons (see Materials and Methods for a more detailed description).

To further analyze the connectivity of these neurons, we looked for the pre- and postsynaptic organization of SELK neurons. We only considered those synaptic partners with ≥10 synaptic connections to limit possible false positive synaptic connections. Most of the postsynaptic partners of SELK neurons (57) localized in the SEZ, as revealed by the BrainCircuits analysis⁵² (**Figure 6B**), with significant projections to the pars intercerebralis. This region contains Insulin Producing Cells (IPCs) that project to the tritocerebrum in the SEZ, where a significant concentration of fibers can be appreciated in the *trans*-Tango immunohistochemistry (**Figure 6C**). This data supports previous results showing that IPCs express the *Lk* receptor (*Lkr*) and that they may be modulated by Lk-expressing neurons^{40,44}. We found five bitter (GRN R1#4, GRN R1#5, GRN R1#12, GRN R2#1 and GRN R2#2) and three sweet (GRN R4#17, GRN R5#12 and GRN R5#14) GRN as presynaptic candidates to GNG.685 (right SELK neuron), and only one bitter (GRN R1#1) and one sweet (GRN R4#11) GRN as presynaptic candidates to GNG.595 (left SELK neuron) (IDs correspond to GRNs described in Engert et al., 2022²⁴; Virtual Fly Brain). That so few presynaptic GRNs were found indicates that SELK proofreading is still required for a much more precise assembly and annotation of the FAFB dataset. We identified a total of 101 presynaptic

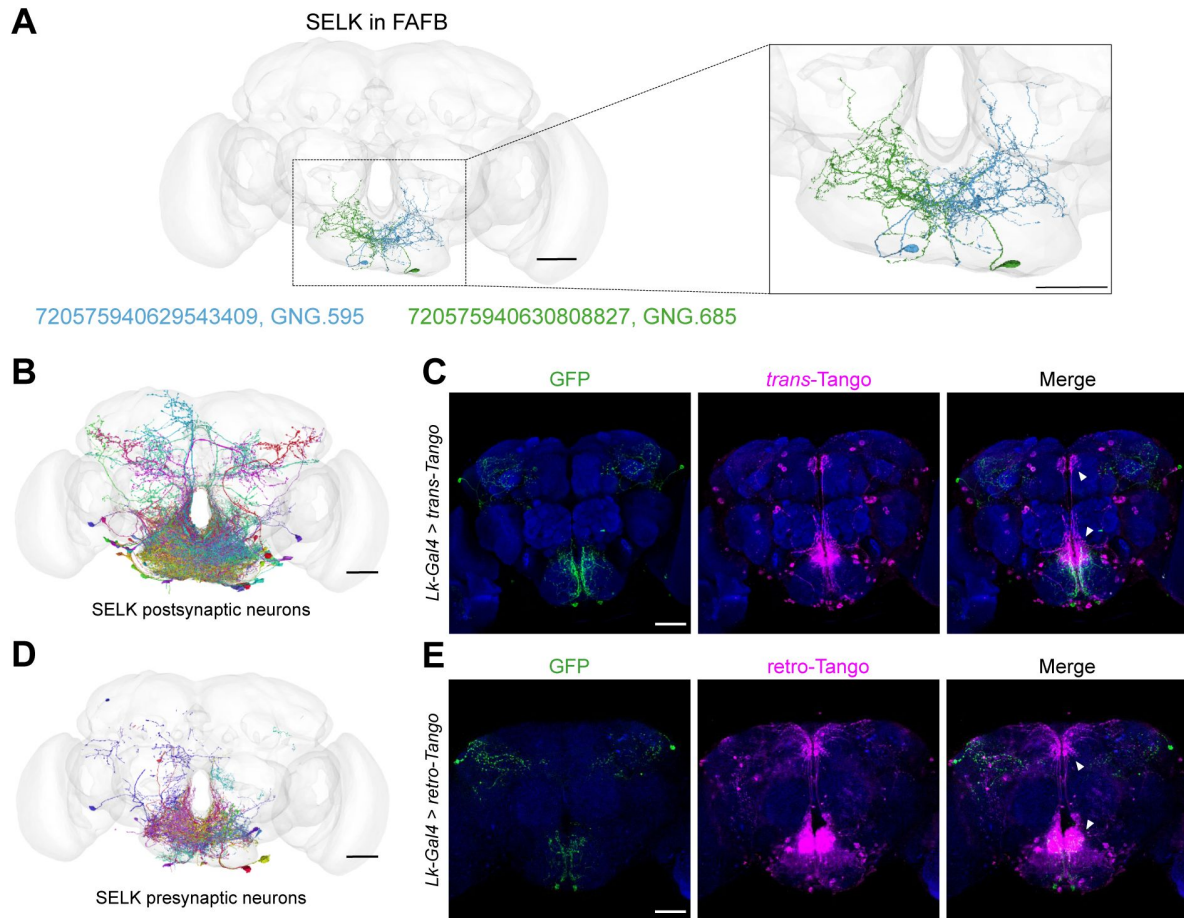


Figure 6.

Lk neurons are pre- and postsynaptic to a significant number of neurons.

(A) The anatomy of the SELK neurons (Right in green and Left in blue) was reconstructed using the FAFB brainmesh template in Flywire. IDs from Flywire are indicated: 720575940629543409 ID for the Left SELK in blue (GNG.595/DNg68(L)) and 720575940630808827 for the Right SELK in green (GNG.685/DNg68(R)). Zoom in to the SEZ from (A), showing the details of the neuronal arborizations. (B) SELK postsynaptic candidate neurons reconstructed in the FAFB (Full Adult Fly Brain) brainmesh template in Flywire. Only postsynaptic candidate neurons with ≥ 10 synaptic points with SELK neurons are shown (57 postsynaptic candidate neurons). (C) Immunofluorescence with anti-GFP (green), anti-RFP (magenta), and anti-nc82 (blue) on a whole-mount brain on a *Lk-Gal4 > trans-Tango* flies. (D) SELK presynaptic candidate neurons reconstructed in the FAFB brainmesh template in Flywire. Only presynaptic candidate neurons with ≥ 10 synaptic points with SELK neurons are shown (101 presynaptic candidate neurons). (E) Immunofluorescence with anti-GFP (green), anti-RFP (magenta), and anti-nc82 (blue) on a whole-mount brain on a *Lk-Gal4 > retro-Tango* flies. Scale bar: 50 μ m.

partners using BrainCircuits with a complex distribution over the brain (**Figure 6D**), though most of the connectivity was found within the SEZ. Additionally, we used *retro-Tango*⁵³, a retrograde trans-synaptic technique that employs similar trans-synaptic mechanisms as *trans-Tango* but in the retrograde direction. This experiment showed that SELK neurons receive most of the presynaptic input from neurons located in the SEZ (**Figure 6E**), consistent with the data derived from the FAFB analysis. Encouragingly, we observed a significant concentration of synapses at the tritocerebrum, similar to the *trans-Tango* signal.

Leucokinin modulates feeding behavior

We have shown that Lk neurons receive direct input from gustatory receptor neurons (GRNs) for bitter (Gr66a^{GRNs}) and sweet (Gr64f^{GRNs}), with Lk neuropeptide expression influenced by the animal's starvation level. Given that Gr64f^{GRNs} facilitate feeding while Gr66a^{GRNs} inhibit it, we sought to understand how Lk neurons integrate gustatory and metabolic signals to modulate feeding behavior. Previous studies indicated that silencing Lk neurons impaired flies' responses to sucrose in a Proboscis Extension Response (PER) experiment⁴⁰. We silenced all Lk neurons in *Drosophila* using the *Lk-Gal4* driver to express the tetanus toxin (TNT) using the *UAS-TNT* driver alongside *UAS-TNT^{imp}* controls⁵⁴, then assessed their response in a PER assay (**Figure 7A**). The light chain of the tetanus toxin expressed with the *UAS-TNT* transgene, blocks all chemical transmission by cleaving the presynaptic protein nSyb. Unexpectedly, silencing Lk neurons did not change the gustatory response to varying sucrose concentrations in 24-hour starved flies (**Figure 7B**). Discrepancies with previous findings may stem from differences in handling and stimuli methodologies⁴⁰.

We hypothesized that Lk neurons might be important for processing sweet and bitter information. To test their role in the PER response to sweet food laced with bitter compounds, we combined 50 mM sucrose with increasing caffeine levels. It is known that *Drosophila* PER to a certain sucrose concentration decreases proportionally to the amount of a bitter compound (i.e. caffeine) it contains² (**Figure 7A**, Experiment 2). The PER experiments included fed, 12-hour, and 24-hour starved flies (**Figure 7C**). Fed flies showed a minimal response to 50 mM sucrose, with caffeine reducing this response consistently across both control and experimental groups. Twelve-hour starved flies exhibited stronger reactions to sucrose than fed flies, with caffeine diminishing PER in a concentration-dependent manner. Twenty four-hour starved flies showed larger PER values that decreased proportionally to the concentration of caffeine. However, this decrease was significantly larger when Lk neurons were silenced using TNT, indicating that Lk neurons are needed to tolerate and feed on food containing bitter, and potentially toxic, compounds. To restrict *Lk-Gal4* transgene expression, we employed the *tsh-Gal80*⁵⁵ construct, which is expressed in the Ventral Nerve Cord (VNC) to restrict Gal4 expression in this region. Restricting TNT expression to SELK and LHLK neurons, did not alter the previously observed PER responses to sweet and bitter compounds (Sup. Figure 7A). To investigate the role of Leucokinin further, we employed an RNAi line designed to knock-down Leucokinin production in Lk-expressing neurons. The results presented in Sup. Figure 7B demonstrate that knocking-down Leucokinin significantly diminishes the flies' tolerance to caffeine in sweet food, indicating that this neuropeptide might be involved in the integration of sweet and bitter signals in Lk neurons.

Finally, we examined whether Lk neurons influence sustained feeding using the flyPAD⁵⁶ to monitor feeding behavior over one hour. We designed two feeding assays (**Figure 7D**). In Experiment 1, flies chose between 20 mM and 100 mM sucrose food. Typically, flies prefer higher sucrose concentrations, but if the higher concentration is mixed with a bitter compound (e.g., caffeine), acceptance of the lower concentration correlates with caffeine levels. Flies with silenced Lk neurons slightly preferred the higher sucrose concentration to controls (**Figure 7E**), yet caffeine addition did not affect their choices. The ingestion amounts for each fly remained comparable in both scenarios. Our findings indicate that Lk neurons are needed to tolerate and initiate feeding of food laced with bitter compounds; however, their role in sustained feeding remains unclear.

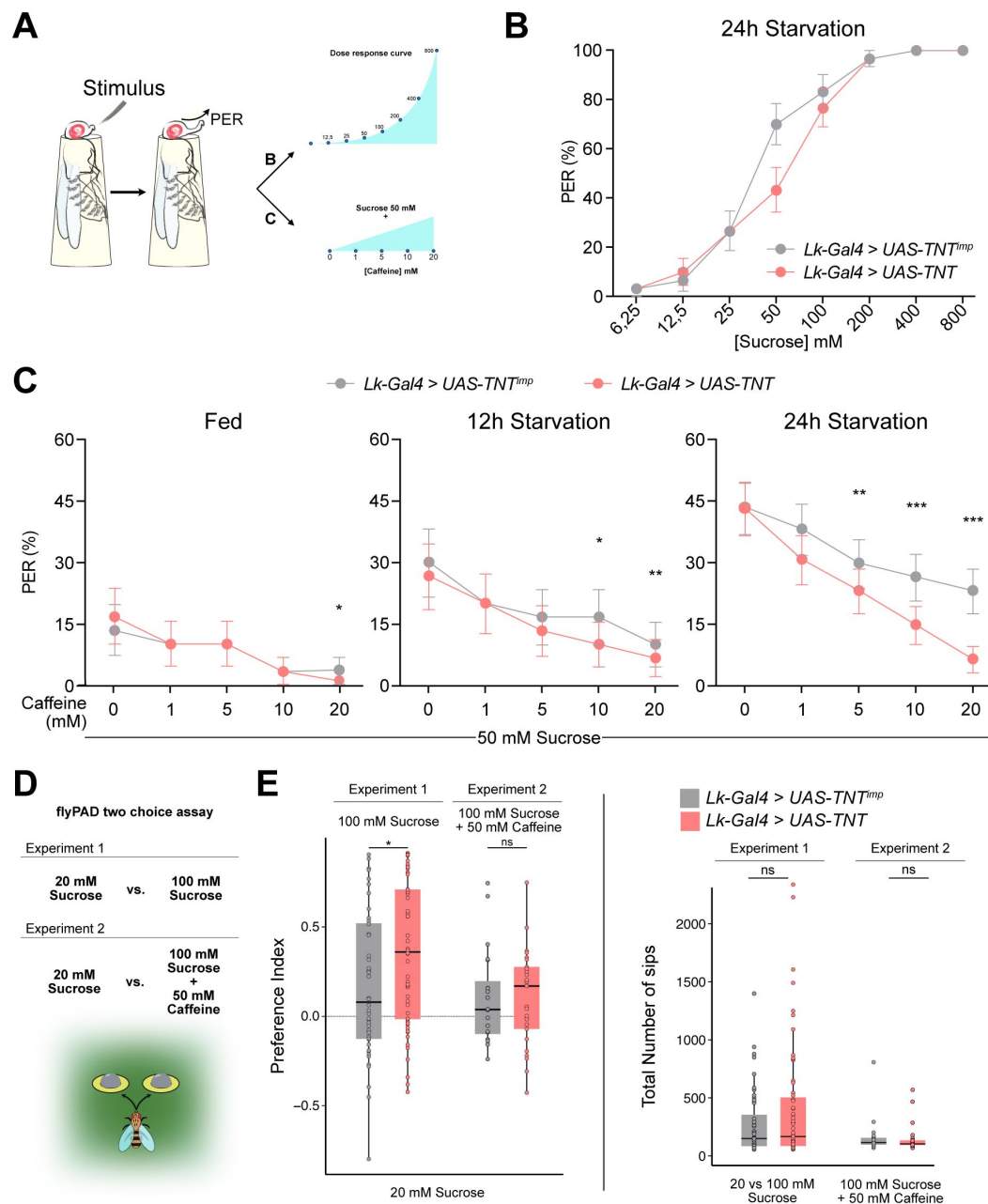


Figure 7.

Leucokinin neurons integrate sweet and bitter gustatory information.

(A) Scheme depicting the experimental procedure for Proboscis Extension Reflex (PER). Flies are given a series of increasing concentrations of sucrose (6.25, 12.5, 25, 50, 100, 200, 400 and 800 mM) for Experiment 1; and a fixed concentration of sucrose (50 mM) mixed with increasing concentrations of caffeine (1, 5, 10 and 20 mM) for Experiment 2. All flies starved 24 hours. (B) PER dose response curve of *Lk-Gal4 > UAS-TNT^{imp}* (n=30) and *Lk-Gal4 > UAS-TNT* (n=30) flies for Experiment 1. (C) PER of *Lk-Gal4 > UAS-TNT^{imp}* (n=30) and *Lk-Gal4 > UAS-TNT* (n=30) flies for Experiment 2. (D) Scheme depicting the experimental procedure for flyPAD (Adapted from Itskov et al., 2014⁵⁶). Experiment 3: 20 mM Sucrose vs. 100 mM Sucrose and Experiment 4: 20 mM Sucrose vs. 100 mM Sucrose + 50 mM Caffeine. All flies starved 24 hours. (E) Preference Index (Left) for the highest concentration of sucrose (100 mM) with or without 50 mM caffeine and total number of sips (Right) for Experiment 3 and 4. Experiment 3: *Lk-Gal4 > UAS-TNT^{imp}* (n=48); *Lk-Gal4 > UAS-TNT* (n=59); Experiment 4: *Lk-Gal4 > UAS-TNT^{imp}* (n=20); *Lk-Gal4 > UAS-TNT* (n=25). Only flies that performed a minimum of 25 sips per fly and 2 bouts were considered for the analysis. ns: non-significant, *p<0.05. PER analysis: Logistic regression model with binomial distribution; Error bars represent the standard error of the proportion. flyPAD statistics Wilcoxon Rank Sum Test.

Discussion

We have demonstrated that Gustatory Second-Order Neurons (G2Ns), which transduce opposing behaviors such as sweet attraction and bitter repulsion, exhibit molecular differences and undergo transcriptomic changes upon starvation. Molecular, connectomic, and behavioral experiments have identified SELK neurons as particularly dynamic G2Ns when under starved conditions. Surprisingly, we discovered that these neurons simultaneously collect information from sweet-sensing Gr64f^{GRNs} and bitter-sensing Gr66a^{GRNs}. Additionally, SELK neurons respond to starvation by increasing the neuropeptide Leucokinin (Lk) expression. These findings reveal that Leucokinin neurons play a novel role in modulating feeding initiation during starvation when flies are presented with sweet food laced with bitter compounds.

Our RNAseq data analysis revealed that SELK neurons only received input from Gr66a^{GRNs}. However, further analysis using immunohistochemistry (*GRN-Gal4 > trans-Tango* + Lk immunohistochemistry) combined with GRASP, active-GRASP and BacTrace, demonstrated that SELK neurons are G2Ns to Gr64f^{GRNs} and Gr66a^{GRNs}. To our knowledge, this is the first demonstration that a pair of G2Ns are broadly tuned to both bitter and sweet gustatory inputs while also integrating metabolic information (**Figure 8A**). *trans-Tango* and its variants have been successfully used to study connectivity and functionality in various neuronal circuits, including gustatory, olfactory, visual systems, mushroom body output neurons or clock neurons^{25,57–60}. Recent analysis of the *retro-Tango* tool, showed that a minimum number of synaptic connections between neurons is needed to label the postsynaptic neurons properly. It is possible that Gr64f^{GRNs} make fewer synaptic contacts with SELK neurons than Gr66a^{GRNs}, decreasing the probability of being labeled by *trans-Tango*. Combined with our stringent FACS protocol, could explain why we were unable to capture SELK neurons as Gr64f^{G2Ns} with FACS.

Current understanding suggests that tastant quality is mediated by labeled lines, where gustatory receptor neurons are segregated to respond to specific taste qualities⁶¹. This segregation is thought to persist further upstream in the circuit, as observed in mice, where taste cells respond to specific tastants, and this information remains segregated at the geniculate ganglion⁶², gustatory cortex⁶³, and the amygdala⁶⁴. Similarly, in *Drosophila* GRNs that detect bitter and sweet compounds project their axons to the SEZ in a non-overlapping fashion¹⁰. Whole-brain *in vivo* calcium imaging revealed that most SEZ interneurons respond to single tastants⁶⁵. However, the same study identified neurons that may respond to multiple tastants, like bitter and sweet, though specific neurons and their circuit positions were not detailed. Other SEZ neurons have been shown to integrate sweet and bitter inputs, such as the E49 motor neuron, which is stimulated by sweet input (Gr5a) and inhibited by bitter input (Gr66a)⁶⁶, but these are not G2Ns. We attempted *in vivo* calcium imaging to further study the gustatory integration in those neurons, but we were unsuccessful due to the challenging location of SELK somas in the medio-ventral region of the SEZ, which is obscured by the proboscis even when fully extended. Additionally, variability in labeling using *UAS-GCaMP7s*, *UAS-GCaMP7m* and *UAS-GCaMP6f* lines hampered the proper imaging of these neurons. Our behavior experiments show that Leucokinin neurons (most likely SELK neurons) are early integrators of gustatory and metabolic information.

In summary, our study highlights the complex integration of gustatory and metabolic information by Leucokinin neurons, providing new insights into the neural mechanisms underlying feeding behavior in *Drosophila*. This work advances our understanding of how sensory and metabolic cues are integrated into the brain to regulate vital behaviors such as feeding.

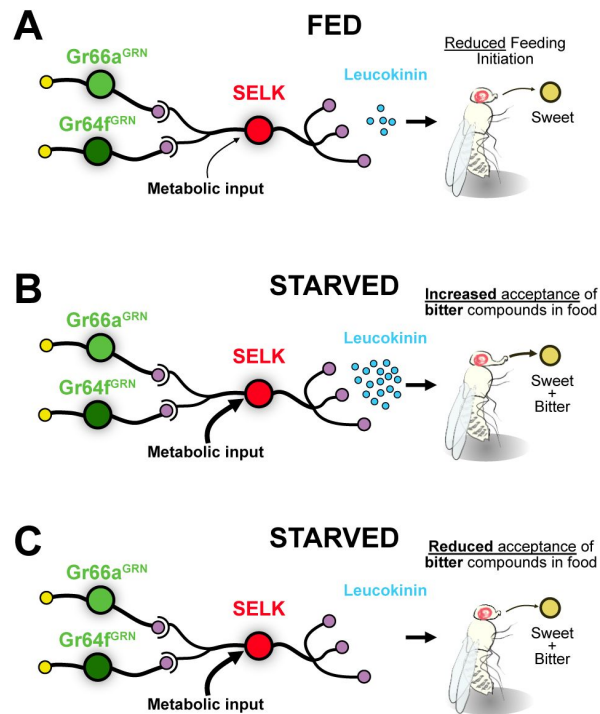


Figure 8.

Model for the integration of gustatory information by SELK neurons.

(A) SELK neurons receive direct input from Gr64f^{GRNs} (sweet), Gr66a^{GRNs} (bitter), and metabolic information. When flies are fed, the expression of the neuropeptide Leucokinin is reduced. The probability of a fed fly initiating feeding is reduced when facing any food. (B) SELK neurons in starved flies express large amounts of Leucokinin. In this situation, flies are more prone to accept food containing bitter compounds. (C) Starved flies with silenced SELK neurons are less prone to accept sweet foods laced with bitter compounds.

Materials and methods

Key Resource Table

<i>Drosophila</i> strains	Source		Identifier
<i>Oregon-R</i>	Kindly donated by Prof. Richard Benton		NA
<i>w¹¹¹⁸</i>	Kindly donated by Prof. Francisco Tejedor		NA
<i>Gr64f-Gal4</i>	Kindly donated by Prof. Richard Benton		NA
<i>Gr64f-LexA</i>	Kindly donated by Prof. Richard Benton		NA
<i>Gr66a-Gal4</i>	Bloomington <i>Drosophila</i> Stock Center		BDSC 57670
<i>Gr66a-LexA; LexAop-rCD2::GFP</i>	Kindly donated by Prof. Richard Benton		NA
<i>Lk-Gal4 (2nd Chr)</i>	Bloomington <i>Drosophila</i> Stock Center		BDSC 51993
<i>Lk-Gal4 (X Chr) used only in tsh-Gal80 experiment</i>	Bloomington <i>Drosophila</i> Stock Center		BDSC 51992
<i>QUAS-mtdTomato; retro-Tango; UAS-retro-Tango-GFP (retro-Tango)</i>	Bloomington <i>Drosophila</i> Stock Center		BDSC 99661
<i>UAS-mCD8::GFP</i>	Bloomington <i>Drosophila</i> Stock Center		BDSC 32186
<i>UAS-myrGFP, QUAS-mtdTomato; trans-Tango (trans-Tango)</i>	Bloomington <i>Drosophila</i> Stock Center		BDSC 77124
<i>UAS-TNT</i>	Bloomington <i>Drosophila</i> Stock Center		BDSC 28837
<i>UAS-TNT^{imp}</i>	Bloomington <i>Drosophila</i> Stock Center		BDSC 28839
<i>UAS-RNAi^{Lk}</i>	Bloomington <i>Drosophila</i> Stock Center		BDSC 25798
<i>UAS-CD4-spGFP1-10, lexAop-CD4-spGFP11 (GRASP)</i>	Bloomington <i>Drosophila</i> Stock Center		BDSC 58755
<i>lexAop-nSyb-spGFP1-10, UAS-CD4-spGFP11; MKRS/TM6B (Active GRASP)</i>	Bloomington <i>Drosophila</i> Stock Center		BDSC 64315
<i>LexAop-GFP, LexAop-QF; UAS-B3RT, QUAS-mtdTomato (BAcTrace)</i>	Bloomington <i>Drosophila</i> Stock Center		BDSC 90826
<i>tsh-Gal80</i>	Kindly donated by Prof. María Domínguez		
Primary antibodies	Host species	Source	Working dilution
Green Fluorescent Protein (GFP)	Chicken	Abcam (ab13970)	1:500
Red Fluorescent Protein (RFP)	Rabbit	Abcam (ab62341)	1:500
nc82 Bruchpilot	Mouse	DSHB	1:50
GFP reconstructed in GRASP	Mouse	Sigma-Aldrich (G6539)	1:500
Leucokinin	Rat	Kindly donated by Prof. Pilar Herrero	1:100

<i>Cadherin DN-Extracellular Domain</i>	Rat	Jackson Immunoresearch 112-165-003	1:500
Secondary Antibodies	Host species	Source	Working dilution
Anti-Chicken Alexa 488	Goat	Abcam (ab150169)	1:250
Anti-Rabbit Cy3	Goat	Jackson Immunoresearch 111-165-144	1:250
Anti-Mouse Cy5	Goat	Jackson Immunoresearch 115-175-146	1:250
Anti-Mouse Cy3	Goat	Jackson Immunoresearch 115-165-166	1:250
Anti-Mouse Alexa 488	Goat	Jackson Immunoresearch 115-545-003	1:250
Anti-Rat Cy3	Goat	Jackson Immunoresearch 112-165-167	1:250
Anti-Rat Alexa 697	Goat	Jackson Immunoresearch 112-605-167	1:250
Chemicals	Source	Reference	
Sucrose	Sigma Aldrich	102174662	
Caffeine	Sigma Aldrich	102143502	
Agarose	Condalab	8100.00	
Agarose (low melting)	Lonza	50100	

Drosophila husbandry

Drosophila melanogaster stocks were reared and maintained on standard “Iberian” fly food under a 12 h light:12 h dark cycle at 25 °C. *w*¹¹¹⁸ and *Oregon-R* fly lines were used as mutant and wild-type strain controls unless otherwise indicated. Starvation was induced by transferring flies into vials containing a solid medium formed with a piece of paper (Kimberly-Clark Kimtech, ref. 7552) soaked with 2,5 ml tap water. All *Drosophila* strains used in this thesis are listed in the Key Resource Table.

Tissue dissection and Fluorescence Activated Cell Sorting (FACS)

Central Brains or SEZs were dissected in cold Calcium- and Magnesium-free 1X Dulbecco’s Phosphate Buffered Saline (DPBS) (ThermoFischer, ref. 14190094), transferred to a Protein LoBind (Eppendorf, ref. 0030108116) tube containing DPBS 0.01% BSA (Bovine Serum Albumin) (Invitrogen, ref. AM2616) and digested with 1 mg/ml of papain and collagenase (Sigma-Aldrich, ref. P4762 and C2674, respectively). After enzymatic digestion, dissociated tissue solution was filtered through a 20 µm filter (pluriSelect, ref. SKU 43-10020-40) into 300 µl DPBS BSA 0.01% in a Protein LoBind tube. Samples were kept at -80°C or directly sorted by FACS.

Cells were sorted using a FACS Aria™ III. Several steps were followed to discard debris, clustered cells and dead cells using a DAPI fluorescent filter. Gustatory Second Order Neurons labeled with mtdTomato were selected using a 555 nm laser (RFP PE-Texas Red A) to discard non-fluorescent cells from the red fluorescent population. In addition to the 555 nm laser, to differentiate those autofluorescence (far red, $\lambda > 670$ nm) from actual mtdTomato *trans*-Tango signal from gustatory second-order neurons ($\lambda = 615$ nm), a far red 566 nm laser (PE-Cy7-A) was applied consecutively to only conserve those mtdTomato fluorescent single cells for PE-Texas Red-A laser and not for PE-Cy7-A laser. All fluorescent gustatory second order neurons were sorted into Protein LowBind

RNase-free tubes containing 15 µl of lysis buffer for RNA extraction and finally stored at -80°C. To sort the Lk GFP⁺ neurons the 488 nm laser was used to sort only GFP⁺ fluorescent cells by applying the FITC-A filter.

RNAseq analysis

RNA extraction and sequencing

RNA was extracted using the PicoArcturus RNA isolation kit (ThermoFisher, ref. 12204-01). The spectrophotometer NanoDrop® ND-1000 was used for samples estimated to contain higher amounts of total mRNA extracted, in the range of 100-3.000 ng/µl. Additionally, to obtain the exact concentration and test the mRNA's quality, total mRNA was measured using the 2100 Bioanalyzer (Agilent Technologies). 50 to 5.000 pg/µl range was measured using chips from the RNA 6.000 Pico Kit (Agilent Technologies).

Samples were sequenced using 50 bp single reads on an *Illumina HiSeq2500* (for fed condition) and *Illumina NextSeq2000* (for starved condition) sequencers at the Genomics Unit of the Center for Genomic Regulation (Barcelona, Spain).

Reads were quality-checked with MultiQC v1.0 software using the tool FastQC v0.11.9. RNAseq output reads were aligned to the *D. melanogaster* reference genome from the Ensemble Project database (EMBL-EBI, Cambridge, UK) using STAR v2.7.9a (Spliced Transcripts Alignment to a Reference)⁶⁷ [↗](#). Differential gene expression analysis was performed using DESeq2 v1.28.1⁶⁸ [↗](#) with the free R programming language (GNU project) software RStudio v4.2. Also, the reads within the gene were transformed by this tool to a total of counts per gene. Transcripts per million (TPM) were calculated using Salmon 1.10.2⁶⁹ [↗](#). A combination of packages (or libraries) from CRAN and Bioconductor v3.14⁷⁰ [↗](#),⁷¹ [↗](#) open-source projects were used for data processing and plotting.

Gene Ontology analysis

The Gene Ontology (GO) analysis was performed using the open source interface PANGEA (Pathway, Network and Gene-set Enrichment Analysis; <https://www.flyrnai.org/tools/pangea/> [↗](#))⁷² [↗](#) specifically the *Drosophila* GO subsets terms at the online platform. Only those genes significantly up-regulated or significantly down-regulated in the differential gene expression analysis were included in the GO analysis. Each of the G2N's differential expressed genes was analyzed separately, and then, only those GO terms of interest were plotted in one single graph.

Additionally, other GO analysis interfaces and tools as g::Profiler (<https://biit.cs.ut.ee/gprofiler/gost> [↗](#)), AmiGO2 (<https://amigo.geneontology.org/amigo> [↗](#)) undefined were used to validate the results of PANGEA. Finally, Gene set enrichment analysis (GSEA) was represented using RStudio v4.2 software (code available upon request).

qPCR

Standard real-time quantitative PCR (qPCR) was performed with 2 ng of template cDNA, PowerUp SYBR Green PCR Master Mix (Applied Biosystems, ref. A25742) and gene-specific primers read on QuantStudio 3 Real-Time PCR System (Applied Biosystems, ref. A28567) with a standard cycling mode: 2 minutes at 50°C and another 2 minutes at 95°C followed by 40x cycles of 15 seconds at 95°C and 1 minutes at 60°C. *Gapdh2* (Glyceraldehyde 3 phosphate dehydrogenase 2) of *D. melanogaster* protein expression levels was used as a housekeeping gene to normalize the results. Triplicate samples per each condition and technical triplicates were performed, and the relative gene expression was normalized by ΔCt analysis. Data is presented as mean ± SEM, and was statistically analyzed using two-tailed Student's *t*-test, considering *p*<0.05 to be statistically significant. Primers used:

Gene	Forward sequence (5' to 3')	Reverse sequence (5' to 3')
<i>LK</i> ⁴⁰	GCCTTTGGCCGTCAAGTCTA	TGAACCTGCGGTACTTGGAG
<i>Gapdh</i> ⁷⁴	CTACCTGTTCAAGTTCGATTTCGAC	AGTGGACTCCACGATGTATTC

Immunohistochemistry

Immunofluorescence on peripheral and central tissues of adult flies was performed following standard procedures^{75,76}. In brief, brains and proboscis were dissected in cold Phosphate Buffer (PB) and fixed for 25 minutes in 4% Paraformaldehyde (PFA EMS15710) in PB at RT. After that, brains and proboscis were washed 5 times for 10 minutes with PB + 0.3% Triton X-100 (PBT) and blocked for 1 hour in PBT + 5% normal goat serum (NGS) (Abcam, ref. ab7481). Later, primary and secondary antibody (see Key Resource Table) incubations were for 48 hours each in PBT + 5% NGS at 4°C in constant agitation. Finally, after 5 washes of 10 minutes, brains and proboscis were equilibrated and mounted in Vectashield Antifade Mounting Medium with DAPI (Vector Laboratories, ref. H-1200-10), maintaining their three-dimensional structure.

Active-GRASP stimulation

Sweet GRN stimulation

Pupae were selected and placed in tubes containing the desired stimuli to prevent adult flies from accessing our standard “Iberian” food which includes yeast and sucrose. *Gr64f-LexA, Lk-Gal4 > active-GRASP* pupae were transferred to tubes with a piece of paper (Kimberly-Clark Kimtech, ref. 7552) soaked in tap water (control group, no activation of Gr64f^{GRNs}) or to a vial containing paper saturated with a 200 mM sucrose solution (sucrose stimulation group, activation of Gr64f^{GRNs}). Adults were maintained in each condition for 48 hours at 25°C until dissection.

Bitter GRN stimulation

Gr66a-LexA, Lk-Gal4 > active-GRASP 1 to 2-day-old adults were transferred into two different conditions: food with sucrose (control condition, no activation of Gr66a^{GRNs}) and food with sucrose and caffeine (bitter stimulation group, activation of Gr66a^{GRNs}). Adding sucrose to the food is necessary to induce the feeding of caffeine-containing food, as the bitter taste itself induces repulsion¹⁰. Female flies were maintained for 3-4 days in each condition until dissection.

- Sucrose food: 200 mM sucrose (Sigma Aldrich, ref. 102174662), miliQ water, 0,9% NaCl, 2,7% yeast and 1% agarose.
- Sucrose + caffeine food: 200 mM sucrose, 75 mM caffeine, miliQ water, 0,9% NaCl, 2,7% yeast and 1% agarose.

To enhance the stimulation of sweet and bitter GRNs, prior to dissection, flies were immobilized on P200 tips and the proboscis was exposed to a piece of paper (Kimberly-Clark Kimtech, ref. 7552) containing 1 M Sucrose (for *Gr64f-LexA, Lk-Gal4 > active-GRASP* flies) and 75 mM Caffeine (for *Gr66a-LexA, Lk-Gal4 > active-GRASP* flies) for 1 hour in a humidified chamber. Following this, the flies were dissected according to the immunohistochemistry protocol.

Microscopy image capture and processing

Images were acquired using a Leica SPEII laser scanning confocal microscope. Routinely, images of dimensions 512x512 pixels were acquired using an oil immersion 20x objective in stacks of 1-2 μm. Files were saved in .tiff format and processed with ImageJ (Fiji) open-source image-processing software⁷⁷. For fluorescence quantification, the fluorescent intensity from the Region Of Interest

(ROI) area was analyzed by using the ROI Manager from ImageJ, and data is presented as mean $\pm$ standard error of the mean (SEM) and was statistically analyzed using two-tailed Student's *t*-test, considering $p < 0.05$ to be statistically significant.

Identification of the SELK neurons in the adult brain *Drosophila* connectome

Neurons were reconstructed in a serial section transmission electron volume⁵¹ using FlyWire (<https://flywire.ai/>)^{78–80}. To accomplish this, 52 sweet and bitter GRNs skeletons (.swc format), previously described²⁴, were downloaded from the open-source interface of Virtual Fly Brain (VFB)⁸¹ and aligned to the Full Adult Fly Brain (FAFB) using the JRC2018U reference brain as a template^{51,82,83}. Among these, 19 GRNs were bitter sensing (12 from group 1 (Sup. Figure 6A) and seven from group 2 (Sup. Figure 6B)), while 33 GRNs were sweet sensing (17 from group 4 (Sup. Figure 6C) and 16 from group 5 (Sup. Figure 6D)). Only Gr64f^{GRNs} and Gr66a^{GRNs} from the right hemisphere were used in this analysis due to the larger dataset. All these GRN skeletons were aligned to the recently released FlyWire dataset, which is a dense, machine learning-based reconstruction of over 80,000 FAFB neurons⁷⁹. Using the Flywire Gateway tool, each of the sweet and bitter GRNs analyzed was aligned and reconstructed to the EM dataset in the FlyWire toolbox, with the IDs annotated⁷⁸.

We next used BrainCircuits⁵² to identify all postsynaptic partners to sweet Gr64f^{GRNs} and bitter Gr66a^{GRNs}. The postsynaptic candidates were downloaded without limiting the number of synaptic contacts among neurons. We noted 12594 postsynaptic hits for Gr66a^{GRNs} and 36790 hits for Gr64f^{GRNs}. By intersecting both datasets, we obtained 333 downstream segment IDs as potential G2Ns receiving inputs from both GRNs. We refined our list to 17 candidates through visual inspection, focusing on neurons with projections within the SEZ (Table 2).

In order to identify which of the 17 candidates presented in Table 2 correspond to the SELK neurons, we proceeded to register the SELK neurons in a reference brain template. To achieve this, immunohistochemistry was performed on 7-day-old *Lk-Gal4 > UAS-CD8::GFP* flies using nc82 (to label the brain structure) and GFP (to label the SELK neurons). Images were taken at a resolution of 1024x1024 and aligned to the “JRC2018U” reference brain template⁸² using CMTK software in Fiji (for more details, see Flywire self-guided training). The aligned images were used to create a skeleton of the SELK neurons using the Single Neurite Tracer (SNT) plugin in Fiji (for more details, see Flywire self-guided training), which was then saved in .swc format. Only the most labeled trace of the SELK neuron (including the soma and main axon) was included in the skeleton to avoid false positive tracings. Finally, the SELK neuron skeletons were aligned to the FlyWire dataset⁷⁸ using the Flywire Gateway tool, taking the JRC2018U reference brain⁸² as a template. The resulting point cloud in the FAFB of the FlyWire Sandbox was used to guide a search for nearby neurons to match the possible SELK neuron candidates' IDs previously identified.

Behavior

Proboscis Extension Reflex(PER)

PER to labellar stimulation was assessed following a standard protocol⁸⁴. Flies were anesthetized on ice and individually immobilized on P200 tips, whose narrow end was cut so that only the fly's head could protrude from the opening, leaving the rest of the body, including legs, constrained inside the tip. After flies were recovered in a humidified chamber for 20 minutes, flies were water-satiated before testing *ad libitum*, discarding those that, after 5 minutes, continued extending their proboscis in response to water. Tastants were delivered using a small piece of paper (Kimberly-Clark Kimtech, ref. 7552) and touching very gently the labellum for 2 seconds maximum, leaving a gap of 1 minute between stimulations. sucrose (Sigma Aldrich, ref. 102174662) and caffeine (Sigma-Aldrich, ref. 102143502) were diluted in milliQ sterile water

Table 2.

List of candidate IDs

Flywire IDs	Name
720575940604891360	
720575940637742150	
720575940624273149	
720575940624253991	
720575940635642725	
720575940633198284	
720575940647033396	
720575940635179871	
720575940606866377	
720575940631349335	
720575940629913130	
720575940632524559	
720575940613670893	
720575940630808827	GNG.685
720575940616353675	
720575940623476407	
720575940630456028	

appropriately. PER was manually recorded: only full proboscis extensions were counted as PER and registered as 1, considering partial or absent proboscis extension as 0. Finally, flies were tested at the end of the experiment with water as the negative control and 1 M of sucrose as the positive control. Only flies that showed negative PER for water and positive PER for 1 M of sucrose were included in the analysis.

flyPAD two choice assay

flyPAD assays were performed to study the feeding microstructure in a two-choice feeding paradigm as described previously⁵⁶, and several feeding parameters were measured individually in a high throughput manner. Individual flies were placed in individual arenas with two different food sources in independent well electrodes. The flyPAD hardware used comprised 56 individual chambers divided into two independent pieces of hardware, each consisting of 28 chambers connected to an independent computer. All tastants used were solved in water 1% low melting agarose (Lonza, ref. 50100). To transfer flies to each chamber, flies were anesthetized in ice for 5 minutes and individually placed in the arena by mouth aspiration according to its sex and genotype. All the experiments began once all flies woke up and were active. Flies were allowed to feed at 25°C for 60 minutes. After time ended, dead flies were annotated. flyPAD data were acquired using the Bonsai framework, and analyzed in MATLAB using custom-written software delivered by *Itskov et al., 2014*⁵⁶. Then, specific R software scripts developed by the lab were applied to analyze the bulk of flyPAD experiment data. Only those flies that performed more than 2 bouts and 25 sips were included in the analysis.

Statistical analysis

The sample size was determined based on preliminary experiments. Data were analyzed using R software v4.1.0 (R Foundation for Statistical Computing, Vienna, Austria, 2005; <https://www.r-project.org>) (code available upon request) and plotted using R or *Graphpad Prism* 6. The statistical test is determined in the figure legend for each of the plots. The Bonferroni method was used when p-value correction for multiple comparisons was required. Except for PER and flyPAD experiments, quantitative data show their distribution by superimposing a boxplot. For the boxplots the whiskers are calculated as follows: the upper whisker equals the third quartile plus 1.5× the interquartile range (IQR) and the lower whisker equals the first quartile minus 1.5× the IQR. Any data points above the superior or below the inferior whisker values are considered outliers. The outliers were included in the statistical comparisons as we performed non-parametric rank tests.

For PER experiments, data were analyzed using a generalized linear model based on a logistic regression set to a binomial distribution model (function `glm()` in R software), where the response of the flies is dependent of their genotype responding to different stimuli of sucrose and caffeine (`glm()` equation used below). Error bars represent the standard error of the proportion ($\sqrt{p(1-p)/n}$).

`glm(Response ~ Genotype * Stimulus, data = PERdata, family = binomial)`

For flyPAD experiments, a set of R software scripts developed by the lab was used to analyze the feeding microstructure behavior, which employs different statistical tests depending on the final goal of the analysis. The preference index is calculated as follows:

$$\text{Preference Index (PI)} = \frac{N_A - N_B}{N_{\text{TOTAL}}}$$

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Additional information

Author contribution

JSA conceived the project and wrote the manuscript with RMA. JSA and RMA performed the brain dissections for the RNAseq analysis. RMA did the FACS sorting, RNA extraction, and sample preparation for sequencing. JSA performed the RNAseq analysis. RMA did the immunohistochemistry, connectomic analysis and PER behavior. MJC performed the flyPAD experiments.

Additional files

Supplementary Figures [↗](#)

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Reviewer #1 (Public review):

Summary:

Mollá-Albaladejo et al. investigate the neurons downstream of GR64f and Gr66a, called G2Ns. They identify downstream neurons using trans-Tango labeling with RFP and then perform bulk RNA-seq on the RFP-sorted cells. Gene expression is up- or downregulated between the cell populations and between fed and starved states. They specifically identify Leucokinin as a neuropeptide that is upregulated in starved Gr66a cells. Leucokinin cells, identified by a GAL4 line, indeed show higher expression when starved, especially in the SEZ. Furthermore, Leucokinin cells colocalize with the trans-Tango signal from downstream neurons of both GRs. This connection is confirmed with GRASP and active GRASP. According to EM data, Leucokinin cells in the SEZ receive a lot of input and connect to many downstream neurons. In behavior experiments performed with flies lacking Leucokinin neurons, flies show reduced responsiveness to sugar and bitter mixtures when starved. The authors suggest that Leucokinin neurons integrate bitter and sugar tastes and that their output is modified by a hunger state.

Strengths:

The authors use a multitude of tools to identify SELK neurons downstream of taste sensory neurons and as starvation-sensitive cells. This study provides an example of how combining genetic labeling, RNA-seq, and EM analysis can be used to investigate the function of specific neural circuits.

Weaknesses:

The authors now provide more evidence to show a functional connection between sensory neurons and SELK neurons, for example, by using active GRASP, however, different staining methods reveal different connectivity patterns. The authors describe a behavioral phenotype when flies are starved, however, the phenotype can still not clearly be assigned to the SELK neurons.

<https://doi.org/10.7554/eLife.100947.2.sa3>

Reviewer #2 (Public review):**Summary:**

A core task of the brain is processing sensory cues from the environment. The neural mechanisms of how sensory information is transmitted from peripheral sense organs to subsequent being processing in defined brain centers remains an important topic in neuroscience. The taste system hereby assesses the palatability of food by evaluating the chemical composition and nutrient content while integrating the current need of energy by assessing the satiation level of the organism. The current manuscript provides insights into the early circuits gustatory coding using the fruit fly as model. By combining trans-tango and FACS-based bulk RNAseq to assess the target neurons of sweet sensing (using by Gr64f-Gal4) and bitter sensing (using Gr66a-Gal4) in a first set of experiments the authors investigate genes that are differentially expressed or co-expressed in normal and starved conditions. With a focus on neuropeptides and neurotransmitters differential expression in the different conditions were assessed resulting in the identification of Leucokinin as potentially interesting gene. The notion is further supported by RNAseq of Lk-Gal4>mCD8:GFP sorted cells and immunostainings. GRASP and BacTrace experiments further supports that the two Lk expressing cells in the SEZ should indeed be postsynaptic to both type of sensors. Using EM-based connectomics data (based on a previous publication by Engert et al.), the authors also look for downstream targets of the bitter versus sweet gustatory neurons to identify the Lk-neurons. Based on morphology they identify candidates and further depict the potential downstream neurons in the connectome, which appears largely in agreement with GRASP experiments. Finally silencing the Lk-neurons shows an increased PER response in starved flies (when combined with bitter compounds) as well as increased feeding in a FlyPad assay.

Strengths:

Overall this is an intriguing manuscript, which provides insight into the organization of 2nd order gustatory neurons. It specifically provides strong evidence for the Lk-neurons as target of sweet and bitter GRNs and provides evidence for their role in regulating sweet vs bitter based behavioral responses. Particularly the integration of different techniques and datasets in an elegant fashion is a strong side of the manuscript. Moreover to put the known LK-neurons into the context of 2nd order gustatory signalling is strengthening the knowledge about this pathway.

Weaknesses:

I do not see any major weakness in the current manuscript. Novelty is to some degree lessened by the fact, that the RNAseq approach did not identify new neurons but rather put the known LK-neurons as major finding. Similarly the final behavioral section is not very deep and to some degree corroborates the previous publication by the Keene and Nässel labs- that said, the model they propose is indeed novel (but lacks depth in analyses, e.g. there is no physiology that would support the modulation of Lk neurons by either type of GRN). The connectomic section appears a bit out of place and after reading it it's not really clear what one should make of the potential downstream neurons (particularly since the Lk-receptor expression has been previously analyzed); here it might have been interesting to address if/how Lk-neurons may signal directly via a classical neurotransmitter (an information that might be found easily in the adult brain single-cell data).

Comments on the latest version:

I feel all points have been included to a satisfactory degree.

<https://doi.org/10.7554/eLife.100947.2.sa2>

Reviewer #3 (Public review):

Summary:

To make feeding decisions, animals need to process three types of information: positive cues like sweetness, negative cues like bitterness, and internal states such as hunger or satiety. This study aims to identify where the information is integrated in the fruit fly brain. The authors applied RNA sequencing on second-order gustatory neurons responsible for sweet and bitter processing, under fed and starved conditions. The sequencing data reveal significant changes in gene expression across sweet vs. bitter pathways and fed vs. starved states. The authors focus on the neuropeptide Leucokinin (Lk), whose expression is dependent on the starvation state. They identify a pair of neurons, named SELK neurons, which express Lk and receive direct input from both sweet and bitter gustatory neurons. These SELK neurons are ideal candidates to integrate gustatory and internal state information. Behavioral experiments show that blocking these neurons in starved flies alters their tolerance to bitter substances during feeding.

Strengths:

- (1) The study employs a well-designed approach, targeting specific neuronal populations, which is more efficient and precise compared to traditional large-scale genetic screening methods.
- (2) The RNAseq results provide valuable data that can be utilized in future studies to explore other molecules beyond Lk.
- (3) The identification of SELK neurons offers a promising avenue for future research into how these neurons integrate conflicting gustatory signals and internal state information.

Weaknesses:

Unfortunately, due to technical challenges, the authors were unable to directly image the functional activity of SELK neurons.

<https://doi.org/10.7554/eLife.100947.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

Mollá-Albaladejo et al. investigate the neurons downstream of GR64f and Gr66a, called G2Ns. They identify downstream neurons using trans-Tango labeling with RFP and then perform bulk RNA-seq on the RFP-sorted cells. Gene expression is up- or downregulated between the cell populations and between fed and starved states. They specifically identify Leucokinin as a neuropeptide that is upregulated in starved Gr66a cells. Leucokinin cells, identified by a GAL4 line indeed show higher expression when starved, especially in the SEZ. Furthermore, Leucokinin cells colocalize with the transTango signal from downstream neurons of both GRs. This connection is confirmed with GRASP. According to EM data, Leucokinin cells in the SEZ receive a lot of input and connect to many downstream neurons. In behavior experiments performed with flies lacking Leucokinin neurons, flies show reduced responsiveness to sugar and bitter mixtures when starved. The authors suggest that Leucokinin neurons integrate bitter and sugar tastes and that their output is modified by a hunger state.

Strengths:

The authors use a multitude of tools to identify SELK neurons downstream of taste sensory neurons and as starvation-sensitive cells. This study provides an example of how combining genetic labeling, RNA-seq, and EM analysis can be combined to investigate neural circuits.

Weaknesses:

The authors do not show a functional connection between sensory neurons and SELK neurons. Additionally, data from RNA seq, anatomical studies, and EM analysis are sometimes contradictory in terms of connectivity. GRASP signal is not foolproof that cells are synaptically connected.

We appreciate the reviewer's comments. Unfortunately, we have not successfully demonstrated a functional response of SELK neurons using in vivo calcium imaging with UAS-GCaMP7 (we tried f, m, and s versions), primarily due to challenges in obtaining stable signals. We stimulated GRNs using sucrose, caffeine, or a mixture of both, and maybe even if the concentrations were high, they were not enough to induce a response.

Regarding GRASP, we acknowledge its limitations as a standalone technique for establishing genuine synaptic connections between neurons, as some signals may reflect false positives resulting from the mere proximity of the candidate neurons. To strengthen our findings, we complemented these results by demonstrating the positive colocalization of the Leucokinin antibody signal over the *Gr66aGal4>trans-TANGO* and *Gr64f-Gal4>trans-TANGO* (Figure 4), confirming that Leucokinin neurons are indeed postsynaptic to both sweet and bitter GRNs. Moreover, we incorporated BacTrace data to highlight the direct connectivity between sweet and bitter GRNs (now Figure 5E).

In the revised manuscript, we have introduced the active-GRASP technique (Macpherson et al., 2015). In this version of GRASP, the presynaptic half of GFP (GFP 1-10) is fused to synaptobrevin, which becomes accessible in the membrane of the presynaptic neuron within

the synaptic cleft upon presynaptic stimulation (in our case, by stimulating with sucrose sweet Gr64f^{GRNs} and with caffeine the bitter Gr66a^{GRNs}). Utilizing this technique, we successfully demonstrated (see new Figure 5B and 5D) that when presented with water, no signal was detected in the *Gr66a-LexA, Lk-Gal4 > active-GRASP*, or *Gr64f-LexA, Lk-Gal4 > active-GRASP* transgene flies. However, in the presence of caffeine, *Gr66a-LexA, Lk-Gal4 > active-GRASP* transgene flies exhibited a clear signal in the SEZ, and similarly, sucrose presentation to *Gr64f-LexA, Lk-Gal4 > active-GRASP* transgene flies yielded a detectable signal. The results obtained from *active-GRASP* provide additional evidence supporting the connectivity between SELK neurons and both Gr64f^{GRNs} and Gr66a^{GRNs}, further indicating the functional connectivity of the GRNs and SELK neurons.

The authors describe a behavioral phenotype when flies are starved, however, they do not use a specific driver for the described cell type, thus they should also tone down their claims.

We agree with the reviewer that the Lk-Gal4 driver line used labels SELK, LHLK, and ABLK neurons. The behavior examined in this paper, the Proboscis Extension Response (PER), measures the initiation of feeding. Although the neural circuit involved in this behavior is primarily confined to the SEZ where SELK neurons are located, we cannot rule out the possibility that other Lk neurons may also play a role in the process. To restrict expression of the Tetanus Toxin, we have utilized the *tsh-Gal80* (Clyne et al., 2008) transgene in combination with the *Lk-Gal4>UAS-TNT* and *Lk-Gal4>UAS-TNT^{imp}* constructs to prevent the expression of the Tetanus Toxin in ABLK neurons, thereby restricting its expression to the SELK and LHLK neurons in the central brain. The new results (Sup Figure 7A) indicate that ABLK neurons do not play a role in integrating sweet and bitter information. However, we acknowledge the reviewer's point that we are still silencing LHLK neurons, so we have adjusted our claims to align more closely with our data

Generally, the authors do not provide a big advancement to the field and some of the results are contradictory with previous publications.

We believe our work does not contradict previous findings, nor does it invalidate the role of ABLK neurons in water homeostasis or the role of LHLK neurons in regulating sleep via starvation. We provide additional information on the possible role of SELK neurons in integrating gustatory information. The location of SELK neurons in the SEZ suggests that they may play a role in feeding behavior, and we have demonstrated that these neurons are indeed involved in integrating gustatory information to influence feeding decisions. We consider we have contributed by highlighting a new role for the Leucokinin neuropeptide in feeding behavior.

Reviewer #2 (Public review):

Summary:

A core task of the brain is processing sensory cues from the environment. The neural mechanisms of how sensory information is transmitted from peripheral sense organs to subsequent being processing in defined brain centers remain an important topic in neuroscience. The taste system hereby assesses the palatability of food by evaluating the chemical composition and nutrient content while integrating the current need for energy by assessing the satiation level of the organism. The current manuscript provides insights into the early circuits of gustatory coding using the fruit fly as a model. By combining trans-tango and FACS- based bulk RNAseq to assess the target neurons of sweet sensing (using Gr64fGal4) and bitter sensing (using Gr66a-Gal4) in a first set of experiments the authors investigate genes that are differentially expressed or co-expressed in normal and starved conditions. With a focus on neuropeptides and neurotransmitters, different

expressions in the different conditions were assessed resulting in the identification of Leucokinin as a potentially interesting gene. The notion is further supported by RNAseq of Lk- Gal4>mCD8:GFP sorted cells and immunostainings. GRASP and BacTrace experiments further support that the two Lk- expressing cells in the SEZ should indeed be postsynaptic to both types of sensories. Using EM-based connectomics data (based on a previous publication by Engert et al.), the authors also look for downstream targets of the bitter versus sweet gustatory neurons to identify the Lk-neurons. Based on the morphology they identify candidates and further depict the potential downstream neurons in the connectome, which appears largely in agreement with GRASP experiments. Finally silencing the Lk- neurons shows an increased PER response in starved flies (when combined with bitter compounds) as well as increased feeding neurons shows an increased PER response in starved flies (when combined with bitter compounds) as well as increased feeding in a FlyPad assay. Strengths:

Overall this is an intriguing manuscript, which provides insight into the organization of 2nd order gustatory neurons. It specifically provides strong evidence for the Lk-neurons as a target of sweet and bitter GRNs and provides evidence for their role in regulating sweet vs bitter-based behavioral responses. Particularly the integration of different techniques and datasets in an elegant fashion is a strong side of the manuscript. Moreover to put the known LK-neurons into the context of 2nd order gustatory signalling is strengthening the knowledge about this pathway.

Weaknesses:

I do not see any major weakness in the current manuscript. Novelty is to some degree lessened by the fact, that the RNAseq approach did not identify new neurons but rather put the known LK-neurons as major findings. Similarly, the final behavioral section is not very deep and to some degree corroborates the previous publication by the Keene and Nässel labs - that said, the model they propose is indeed novel (but lacks depth in analyses; e.g. there is no physiology that would support the modulation of Lk neurons by either type of GRN). The connectomic section appears a bit out of place and after reading it it's not really clear what one should make of the potential downstream neurons (particularly since the Lk-receptor expression has been previously analyzed); here it might have been interesting to address if/how Lk-neurons may signal directly via a classical neurotransmitter (an information that might be found easily in the adult brain single-cell data).

We thank the reviewer for the comment. Indeed, we attempted *in vivo* Ca imaging but were unsuccessful. We have rewritten the connectomic section to better integrate it with the rest of the text and have reanalyzed the data obtained. We considered gathering data from the single-cell adult dataset, but this dataset includes the entire adult fly brain, encompassing SELK and LHLK neurons, making it impossible to differentiate between the two types of Lk neurons. Any further analysis will require transcriptomic analysis of SELK via scRNAseq under the different metabolic conditions tested in this study work.

Reviewer #3 (Public review):

Summary:

To make feeding decisions, animals need to process three types of information: positive cues like sweetness, negative cues like bitterness, and internal states such as hunger or satiety. This study aims to identify where the information is integrated into the fruit fly brain. The authors applied RNA sequencing on second-order gustatory neurons responsible for sweet and bitter processing, under fed and starved conditions. The sequencing data reveal significant changes in gene expression across sweet vs. bitter pathways and fed vs. starved states. The authors focus on the neuropeptide Leucokinin

(Lk), whose expression is dependent on the starvation state. They identify a pair of neurons, named SELK neurons, which express Lk and receive direct input from both sweet and bitter gustatory neurons. These SELK neurons are ideal candidates to integrate gustatory and internal state information. Behavioral experiments show that blocking these neurons in starved flies alters their tolerance to bitter substances during feeding.

Strengths:

(1) The study employs a well-designed approach, targeting specific neuronal populations, which is more efficient and precise compared to traditional large-scale genetic screening methods.

(2) The RNAseq results provide valuable data that can be utilized in future studies to explore other molecules beyond Lk.

(3) The identification of SELK neurons offers a promising avenue for future research into how these neurons integrate conflicting gustatory signals and internal state information.

Weaknesses:

(1) Unfortunately, due to technical challenges, the authors were unable to directly image the functional activity of SELK neurons.

(2) In the behavioral experiments, tetanus toxin was used to block SELK neurons. Since these neurons may release multiple neurotransmitters or neuropeptides, the results do not specifically demonstrate that Leucokinin (Lk) is the critical factor, as suggested in Figure 8. To address this, I recommend using RNAi to inhibit Lk expression in SELK neurons and comparing the outcomes to wild-type controls via the PER assay.

We appreciate the author's comments and suggestions. As noted, Tetanus Toxin silences the neuron's activity, affecting the functioning of various neurotransmitters and neuropeptides released by the targeted neuron. In response to the reviewer's recommendation, we employed an RNAi line specifically designed to silence Leucokinin production in Lk-expressing neurons.

The results presented in Supplementary Figure 7B demonstrate that knocking down Leucokinin in Lk neurons significantly reduces the flies' tolerance to caffeine in sweet food.

It is crucial to highlight that the sucrose concentration used in Figure 7C was 50mM, whereas in Supplementary Figure 7B, it was increased to 100mM. This adjustment was necessary because the *Lk-Gal4*, *UAS-RNAi*, and *Lk-Gal4>UAS-RNAi* transgenic lines exhibited reduced sensitivity to sucrose compared to the *Lk-Gal4>UAS-TNT* or *Lk-Gal4>UAS-TNT^{imp}* lines. We aimed to establish a sucrose concentration that would elicit a 50% Proboscis Extension Response (PER) without adding any other compound, thereby allowing us to evaluate the additional effect of caffeine in the food.

However, according to the data derived from the connectome, SELK neurons might be cholinergic, and this neurotransmitter might be involved in controlling also the behavior of the flies.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

To get more evidence for connections between sensory cells and SELK neurons, could the authors also analyze a second available EM data set? Would setting a different threshold (>5 synapses) reveal connections to both sensories? Comparisons between SELK in- and outputs from EM data and Tango labeling also seem to differ quite a lot based on

provided images - can the authors count cell bodies in the stainings? Further proof would be to provide functional imaging data that shows that SELK neurons respond to sugar and bitter compounds.

In this study, we utilized the recently published EM dataset for the *Drosophila* central brain connectome (Dorkenwald et al., 2024; Flywire.ai). Changing the number of synapses affects the counts of pre- and postsynaptic neurons. We set a threshold of more than five synapses, as recommended by Flywire, to avoid false positives (Dorkenwald et al., 2024). This threshold has been widely used in recent papers (Engert et al., 2022; Shiu et al., 2022; Walker et al., 2025).

The neuron counts in the connectomic data differ from those in the *trans*- and *retro*-TANGO experiments. In our initial *trans*-TANGO experiment, which labeled postsynaptic neurons in the Gr64fGal4 and Gr66a-Gal4 transgenic lines, we counted the labeled neurons (see Supplementary Figure 1C) and observed considerable variability between different brains. Due to anticipated variability, we did not count the labeled neurons from *trans*-TANGO and *retro*-TANGO techniques in the Leucokinin neurons. Furthermore, neither technique labels all postsynaptic or presynaptic neurons, respectively. A recent study on the *retro*-TANGO technique (Sorkac et al., 2023) found a minimum threshold: the presynaptic neuron must form a certain number of synapses with the neuron of interest to be adequately labeled. According to this paper, the established threshold is 17 synapses. It is likely that the *trans*-TANGO technique also has a threshold relating to the number of labeled neurons, contingent on the synapse count. This would explain the discrepancy between the two results.

Unfortunately, we have not been able to provide functional data pointing to the activation of SELK neurons by sucrose or caffeine. However, our active-GRASP data indicates that the connectivity between Gr64f^{GRNs} and Gr66a^{GRNs} with SELK neurons is present and functional.

How many Leucokinin-positive cells are in the SEZ? Does the RNA-seq data provide further information about the SELK neurons? Potential receptor candidates for how they integrate hunger signals? AMPKa was described to be required in LHLK neurons.

There are two SELK neurons in the SEZ. Due to the nature of our bulk RNA sequencing (RNAseq), we cannot link any additional gene expressions detected in our transcriptomic analysis specifically to the SELK neurons regarding the integration of various signaling processes. Furthermore, the single-cell RNA sequencing (scRNAseq) data available from the *Drosophila* brain, as reported by Li et al. (2022), does not allow accurate differentiation between SELK and LHLK neurons. To understand how these neurons integrate both metabolic and sensory information, it is crucial to conduct a focused RNAseq study specifically on the SELK neurons to understand how these neurons integrate both metabolic and sensory information. This targeted analysis would provide the necessary insights to elucidate their functional roles better. However, according to the data derived from the connectome, SELK neurons might be cholinergic, and this neurotransmitter might be involved in controlling also the behavior of the flies.

According to previous studies (Yurgel et al., 2019), the Lk-GAL4 line is also expressed in the VNC, thus the authors could make use of the tsh-GAL80 tool to clean up the line. This study also performed GCaMP imaging in fed and 24h starved animals in SELK and couldn't find a difference, can the authors explain this discrepancy?

We thank the reviewer for this suggestion. We have now added a new piece of data using the *tsh-Gal80* transgene in our PER experiments (Supplementary Figure 7A). Blocking the expression of TNT in the ABLK neurons does not affect the main conclusion of the behavioral results. As stated previously, we were unable to obtain *in vivo* Ca imaging responses in SELK neurons upon exposure to sucrose, caffeine, or mixtures of sucrose and caffeine. We do not

believe this is a discrepancy with previous works like Yurgel et al., 2019. It is likely that we faced technical issues regarding expression stability and that the stimulation was possibly too weak to detect changes in GFP levels

Reviewer #2 (Recommendations for the authors):

As mentioned above I do not have any major comments on the manuscript, but there are a few points that I feel should be considered:

(1) The identification of the Lk-candidate neurons in the connectome remains a bit mysterious. In the method sections, this reads as follows "manual and visual criteria were applied to identify the neurons of interest ". a) What precisely was done to get to the candidates? b) Are there alternative candidates that may be Lk-neurons? c) How would another neuron affect the conclusion of the downstream analysis?

We thank the reviewer for this comment. We have now modified and added new information in the connectomic section, reinforcing our conclusions and correcting the results obtained.

Our GRASP, BacTRace, and immunohistochemistry experiments pointed to SELK neurons as postsynaptic to both Gr64f^{GRNs} (sweet) and Gr66a^{GRNs} (bitter). To identify which neurons in the connectome could be the SELK neurons, we utilized a previously described set of GRNs already identified in the connectome (Shiu et al., 2022). We extracted all postsynaptic neurons to the sweet and bitter GRNs identified and intersected both datasets, retaining only those candidate hits receiving simultaneous input from sweet and bitter GRNs. This process yielded a total of 333 hits. Through visual inspection, we discarded all hits that were merely neuronal fragments or neurons that clearly were not our candidates. We narrowed the list down to a final set of 17 candidate neurons whose arborization was located in the SEZ. We reduced the candidates to two final entries from this list: ID 720575940623529610 (GNG.276) and ID 720575940630808827 (GNG.685). The GNG.276 neuron had a counterpart in the SEZ identified as GNG.246. Both of these neurons were annotated as DNg70 in the Flywire database. GNG.685 had a counterpart identified as GNG.595, and these two neurons were classified as DNg68. In both cases, the neuronal candidates, DNg70 and DNg68, were classified as descending neurons, a characteristic of previously described SELK neurons (Nässel et al., 2021). In our initial analysis published in bioRxiv and sent for revision, we identified DNg70 as potentially the SELK neurons based solely on the morphology of the neurons via visual inspection. However, we employed a better method to determine which candidate is more likely to be the SELK neurons, concluding that DNg68, rather than DNg70, represents the SELK neurons. Briefly, we performed an immunohistochemistry for GFP in the *Lk-Gal4>UAS-CD8:GFP* flies. We aligned the resulting image in a *Drosophila* reference brain (JRC2018 U) using the CMTK Registration plugin in ImageJ. The resulting image was skeletonized using the Single Neurite Tracer plugin in ImageJ and later uploaded to the Flywire Gateway platform to compare the structure of the aligned and skeletonized SELK neurons to our candidates. This comparison clearly indicated that the DNg68 neurons are the best candidates for representing the SELK neurons, rather than DNg70. We have updated the text and Figures 6 and Supplementary Figure 6 to reflect the new results. These new results do not alter the conclusions of the paper.

(2) In the transcriptomic experiments It seems that the raw transcripts are reporters, rather than normalised data. Why?

All transcriptomic data is normalized. In Figure 1 the differential expression was calculated using Deseq2 normalized counts. In Figure 2, Transcripts Per Million (TPM) were calculated using the Salmon package and normalized for the gene length.

(3) The expression of nAChRbeta1 in the transcriptomic data is rather striking. However, this remains currently not addressed: is this expression real?

We have not confirmed the upregulation or downregulation in gene expression for other but for Leucokinin, which is our main interest. We found the presence of nAChRbeta1 interesting, as GRNs are cholinergic (Jaeger et al., 2018), suggesting that it would make sense to find cholinergic receptors in G2Ns. However, it is possible that these receptors are expressed in all G2Ns and serve as a common means of communication.

(4) The description of the behavioural experiments in the results section is rather brief. I had a hard time following it since the genotypes are not repeated nor is it stated what is different in the experimental group vs control (but instead simply what changes in the experimental group, in a rather discussion-like fashion).

We thank the reviewer for the comment, we have rewritten this section to improve its clarity.

(5) If I understand the genetics for the behavioural experiments correctly it addresses the entire Lk-Gal4 expressing population, thus it is not possible to describe the role of the two SEZ neurons, but rather LkGal4 neurons. This should be clarified.

We thank the reviewer for this comment. Indeed, the Lk-Gal4 driver we used drives expression in all Leucokinin neurons, making it impossible to distinguish between the SELK, LHLK, or ABLK neurons. We have added a new piece of behavioral data by using the tsh-Gal80 transgene to prevent the expression of TNT in the ABLK neurons (Supplementary Figure 7A), but still we cannot distinguish between SELK and LHLK. We have rewritten the text to clarify this fact.

Reviewer #3 (Recommendations for the authors):

Overall, the manuscript is well-written, I only have one minor suggestion for improvement. In Figure 8C, please clarify the use of TNT to block Lk release.

We thank the reviewer for the comment, we have clarified the use of TNT in the text.

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